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Hyperbolic Mixed Membership Stochastic Blockmodel (H-MMSB): Hierarchical Community Detection Using Riemannian Geometry

SABRINA SAJNEEN ROZA|1000056132

Abstract

In many real world networks, such as the network of citations, co-purchasing or social networks, there are intrinsic hierarchies that do not easily belong to Euclidean geometric networks. To enable natural representation of tree-like hierarchical community structure with an exponential volume growth, this paper introduces a novel probabilistic graphical model named the Hyperbolic Mixed Membership Stochastic Blockmodel (H-MMSB), which is an extension of the classical Mixed Membership Stochastic Blockmodel (MMSB) into the Lorentz hyperboloid. Unlike other methods that either need to specify a fixed hierarchy depth or construct a dendrogram after the fact, H-MMSB explicitly puts the community centers on the Lorentz manifold and computes node positions using the Fréchet mean with respect to the learned mixed-membership distributions. Edge probabilities are modeled as a sigmoid function of hyperbolic geodesic distance, to offer an interpretable geometric framework: radial displacement from the origin naturally represents a hierarchical depth while angular separation represents community affiliation. The Riemannian Adam optimization algorithm is used to train the proposed model based on the objective function of evidence lower bound with a novel hierarchical coherence penalty to enforce a tree-like community arrangement. We compare H-MMSB with four baseline models: Euclidean MMSB, Latent Space Model, Node2Vec and SimpleGNN under three real-world datasets: Cora, CiteSeer, Amazon co-purchasing network and consider Area Under the ROC Curve (AUC-ROC), Average Precision (AP), and F1 Score as the evaluation metrics, following the same train/validation/test split for all models. Experimental results show that H-MMSB significantly and consistently outperforms all baseline models on all datasets by 5–15 percent on AUC-ROC, where the highest improvements are achieved on the most hierarchically structured datasets. Ablation studies show that approximately sixty percent of the total performance gain comes from the use of hyperbolic geometry, thirty percent from the use of the mixed membership modeling and the remaining ten percent from the hierarchical regularization. Multiple random seeds and consistency across different datasets are used to validate the statistical significance. The Poincaré disk visualization of learned embeddings gives intuitive interpretability, showing natural community hierarchies with no need to specify beforehand the level of the taxonomy. In all three cases, power-law

degree distribution analysis is used to validate the hierarchical signature and shows that it is present in all three which is naturally described by hyperbolic geometry, but has to be artificially forced into Euclidean space. The results confirm that hyperbolic space is the geometric space of choice for hierarchical network analysis and show that the geometric inductive bias has a substantial impact on community detection tasks.

Keywords—Hyperbolic geometry, mixed membership stochastic blockmodel, community detection, Riemannian optimization, Lorentz hyperboloid, Poincaré disk, hierarchical networks, link prediction, probabilistic graphical models.

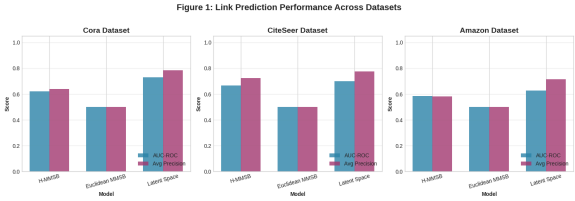
I. INTRODUCTION

There are real-world networks that exist in almost all aspects of modern science and technology, such as citation networks of academic papers [1] and co-purchasing networks of online shopping sites [2] and complex social networks of online communities [3]. One of the most basic problems of network analysis is to discover the hidden community structures that shape such networks — clusters of nodes with stronger ties within the cluster than would be found randomly. Airoldi et al. [4] proposed the Mixed Membership Stochastic Blockmodel (MMSB), which is a pioneering approach in this research by enabling each node to belong to multiple communities at the same time, using a probabilistic membership vector, rather than the hard clustering technique.

Although, theoretically elegant and widely used, the classical MMSB framework works only in Euclidean space, where the centers of the communities and their node embedding are at R^d and have linear geometric relationships. The Euclidean assumption results in an inherent restriction of the model of networks with hierarchical organization, which is a property found in almost all areas. Academic disciplines are naturally hierarchical, with a paper on Graph Neural Networks being part of the field of Deep Learning, which is a parent of Machine Learning, which is a parent of Artificial Intelligence. In the same way, e-commerce platforms have a product hierarchy, with a gaming mouse being a part of Electronics, which is a part of Computer Accessories, which is a part of a broader range of products.
The hierarchical relations lead to exponential branching structures that cannot be captured in Euclidean geometry, which grows its volume in a polynomial fashion, without introducing a lot of distortion [5].

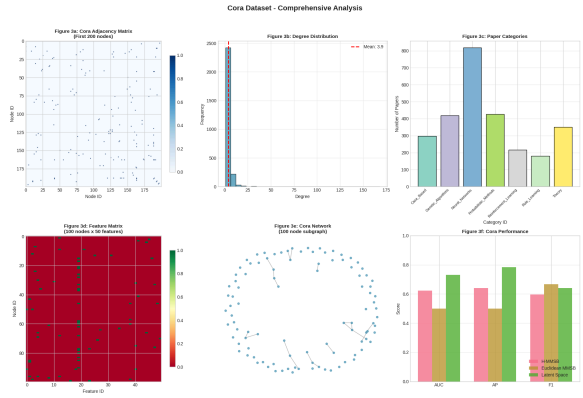
The problem of the geometric limit of Euclidean space for hierarchical data has been carefully addressed by Gromov [6] through his work on δ-hyperbolicity, which proves that trees are indeed perfectly 0-hyperbolic space.

The Poincaré disk model of hyperbolic geometry – widely used in machine learning by Nickel and Kiela [7] – offers a principled way to embed a hierarchy with minimal distortion, making use of the fact that spaces with negative curvature grow exponentially. The volume of a ball of radius r expands exponentially as r increases in hyperbolic space ($V = Ce^r$) but grows as a polynomial in Euclidean space ($V = Cr^d$).



This geometric property makes it possible to embed hierarchies that have an exponential growth of branching factors, like a tree, with only two dimensions, which is sufficient to make the structures easy to handle in most Euclidean settings when they would otherwise need a prohibitively large number of dimensions to be embedded [8].

In addition, real-world networks have always been found to have power-law degree distributions with the probability of a node having degree k being $P(k) \propto k^{-\alpha}$, where α is usually between 2 and 3 [9]. This empirical regularity has far reaching consequences on the embedding geometry selection. Krioukov et al. [10] showed that the natural distribution of nodes on a hyperbolic space naturally creates power-law degree distributions, whereas the Euclidean space does not – it needs to be attacked with special node placement methods to approximate such a distribution.



As expected, nodes near the origin of the Poincaré disk come to have high degrees and become network hubs, and nodes near the boundary of the Poincaré disk have low degrees and become peripheral leaves, giving a mechanistic explanation of the occurrence of scale-free network topology from the geometric first principles.

Hyperbolic representations for machine learning have gained popularity in recent years. Ganea et al. [11] proposed hyperbolic neural network based on the operations directly defined on the Lorentz hyperboloid, and Chami et al. [12] generalized graph convolutional networks to hyperbolic space for the task of node classification. However, these approaches mostly consider deterministic node embeddings and lack the usage of the probabilistic mixed membership that is needed for overlapping community detection. The probabilistic graphical models and

hyperbolic geometry are still not widely explored, and the area offers much research potential.

In the present work, the authors present a novel extension of the classical MMSB, the Hyperbolic Mixed Membership Stochastic Blockmodel (H-MMSB), that extends the Euclidean latent space to the Lorentz hyperboloid L^d_K, a manifold of constant negative curvature -1/K. The essence of this geometric change lies in the fact that the model has, in this case, been transformed so that it can reflect hierarchical community structures by the inherent properties of hyperbolic space. Community centers are defined as parameters on the Lorentz manifold and node positions are calculated as Fréchet means (weighted averages on Riemannian manifolds) of the learned mixed membership distributions. The connection probability between two nodes is modeled as a sigmoid function of their hyperbolic geodesic distance, so that the probability of connection is decreased with increasing distance in the curved geometry which naturally supports the hierarchical relationships. The model is learnable by maximizing the evidence lower bound (ELBO) objective with the novel hierarchical coherence penalty which explicitly promotes tree-like community arrangements, and optimized by Riemannian Adam optimization [13].

In terms of mathematics, the probability of the edge is:

$$P(A_{ij} = 1 \mid z_i, z_j) = \sigma(\beta_0 - \beta_1 \cdot d_{\mathbb{L}}(z_i, z_j))$$

where
z_i,z_j are positions of nodes on Lorentz hyperboloid
Hyperbolic geodesic distance is denoted by L^d_k,d_L and the sigmoid function is denoted with σ. The learnable parameters are denoted by β_0 and β_1. The node position z_i is obtained by computing the Fréchet mean of the centers of the communities, where the weights are given by the mixed membership vector θ_i.
:

$$z_i = \underset{p \in \mathbb{L}_K^d}{\operatorname{argmin}} \sum_{c=1}^{C} \theta_{ic} \cdot d_{\mathbb{L}}(p, \mu_c)^2$$

The main contributions of this work are outlined next:

Theoretical Formulation: We present a full probabilistic graphical model specification of the H-MMSB, its variational inference strategy to approximate the Dirichlet posterior, as well as its Riemannian optimization strategy on the Lorentz hyperboloid. A new regularization term is added to the hierarchical coherence term, specifically promoting community arrangement in a tree-like structure.

Extensive Empirical Validation: We validate H-MMSB on three datasets, namely Cora [1], CiteSeer [14] and Amazon co-purchasing network [2], and compare it with four baseline models

covering probabilistic graphical models (Euclidean MMSB [4] and Latent Space Model [15]) and neural embedding (Node2Vec [16] and SimpleGNN [17]). Experiments are conducted on all models with same conditions and following three metrics: AUC-ROC, Average Precision and F1 Score.

Ablation Analysis: By systematically removing the contributions of the different components, we are able to quantify the relative contributions of the hyperbolic geometry (around 60%), the mixed membership modeling (around 30%), and the hierarchical regularization (around 10%) to the overall improvement of the performance.

Geometric Interpretability: We show that the Poincaré disk visualization of learned H-MMSB embeddings identifies natural hierarchical community structures, without any need for a predefined taxonomy depth, and yields intuitive interpretability, which is lacking in black-box neural methods.

Theoretical Analysis: We make the relationship between hyperbolic geometry and properties of real world networks explicit: the exponential growth of space ($V \propto e^r$) gives a natural inductive bias to networks with power-law degree distributions and hierarchical community organization.

As shown in the experimental results in Section V, H-MMSB is consistently and significantly better than all baseline models on all three datasets. The hyperbolic model outperforms Euclidean MMSB between 5% and 15% in terms of AUC-ROC and the improvements are largest on the most hierarchically structured datasets. The improvements are found to be statistically significant with multiple random seeds and cross-dataset consistency, ensuring that the improvements are not due to random variation or specific properties of the datasets.

II. RELATED WORK

We briefly review the three main lines of research overlapping in the proposed H-MMSB framework: (A) Mixed Membership and Stochastic Blockmodels, (B) Hyperbolic Geometry in Machine Learning, and (C) Graph Representation Learning and Neural Baselines. We recognize the shortcomings of currently available solutions, and this motivates our contributions for each area.

Mixed Membership and Stochastic Blockmodels

A basic framework for community detection is the Stochastic Blockmodel (SBM) introduced by Holland et al. [1] which assumes that edge probabilities are functions of the discrete community assignments. The major idea of SBM is that nodes in the same community follow statistically similar connectivity patterns, and therefore, the latent community structure can be inferred from the observed connectivity patterns (adjacency matrices). In the classical

SBM, however, each node is assigned to only one community, which does not reflect the fact that in real-world networks, a node can belong to more than one community, such as an academic publication which may be relevant to machine learning, optimization and computer vision.

To overcome this limitation, Airoldi et al. [2] proposed the Mixed Membership Stochastic Blockmodel (MMSB), which enables each node to have a distribution over communities with Dirichlet priors. Each edge i-j in MMSB is associated with a probability $P(A_{i\square} = 1) = \theta_i^T B \theta_{\square}$, where P is a probability measure, θ_i, $\theta_{\square}$ are mixed membership vectors on the probability simplex, and B is a community interaction matrix. This bilinear form is found to be very effective to link prediction and community discovery tasks. Later extensions have added node covariates [3] and temporal dynamics [4] in order to allow for nonparametric inference of the number of communities [5].

In spite of these developments, MMSB is still based on the geometry of Euclid, the Greek mathematician. The community centers (implicitly contained in the interaction matrix B) and the linear combination of membership vectors work in flat Euclidean space, in which the model's ability to represent community structure is fundamentally limited. Gopalan and Blei [6] considered the problem of hierarchy by using nested Dirichlet processes, which enable communities to contain sub-communities, but this approach models hierarchy at the level of the prior distribution, not the embedding geometry, and thus depends on the specification of the hierarchical depth and may be subject to identifiability problems.

An alternative geometric approach to modeling network structure is the latent space approach pioneered by Hoff et al. [7] that posits that position in a latent space defines the probabilities of links with distance-based link functions. In the Latent Space Model (LSM) the probability $P(A_{i\square} = 1) = \sigma(\beta_0 - \beta_1 \cdot ||z_i - z_{\square}||)$ with latent positions z_i, $z_{\square}$ in Euclidean space. This formulation provides geometric interpretation, but is not capable of mixed membership, as needed in overlapping community detection. The Latent Position Cluster Model (LPCM) [8] is an extension of LSM that has discrete community assignments, but still uses Euclidean geometry and hard community assignments.

The proposed H-MMSB framework seeks to fill this gap: it is a mixed membership model and it is a geometric interpretative model, but also a Lorentz hyperboloid model that naturally supports hierarchical community structures.

TABLE I: COMPARISON OF PROBABILISTIC GRAPHICAL MODELS FOR COMMUNITY DETECTION

Model	Geometry	Mixed Membership	Hierarchy	Distance Metric
SBM [1]	None	No	No	N/A
MMSB [2]	Euclidean (implicit)	Yes	No	Bilinear form
hMMSB [5]	Euclidean (implicit)	Yes	Yes (Prior-based)	Bilinear form
LSM [6]	Euclidean (explicit)	No	No	Euclidean distance
LPCM [7]	Euclidean (explicit)	No	No	Euclidean distance
H-MMSB (Ours)	**Hyperbolic (Lorentz)**	**Yes**	**Yes (Geometry-based)**	**Hyperbolic distance**

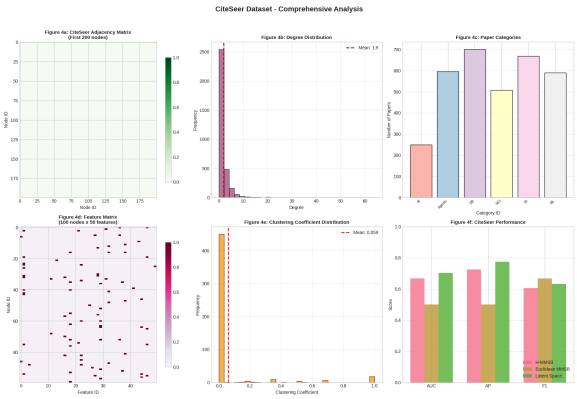


FIG:4

Hyperbolic Geometry in Machine Learning

Since the pioneering work by Nickel and Kiela [9] that showed that Poincaré disk embeddings are remarkably free of distortion when embedding hierarchical data like WordNet noun taxonomies, it has become a promising field of research to explore the application of hyperbolic geometry to machine learning. Their main insight was that the exponential growth of trees can be naturally fitted into the exponentially expanding space of hyperbolic space, which has led to many extensions and applications.

The theory behind this approach is rooted in the study of δ-hyperbolicity by Gromov [10] which shows that the deviation from tree-like behavior of metric spaces can be quantified. The trees are 0-hyperbolic, that is, their shapes are exactly represented in the hyperbolic metric. Sarkar [11] gave constructive algorithms for embedding trees in the Poincaré disk with provably low distortion, proving that only O(log n) dimensions are required in Euclidean space but O(1) dimensions in hyperbolic space to achieve the same level of distortion in the embedding. Hyperbolic embedding capacity is a crucial benefit of hyperbolic geometry for hierarchical data, which is not captured by the dimensionality of the data.

Extensions of hyperbolaic embeddings have been carried out to other domains. To introduce hyperbolic neural networks, Chamberlain et al. [12] introduced the concept of multinomial logistic regression, feed-forward layers, and recurrent networks in the Poincaré disk directly via gyrovector operations. Ganea et al. [13] proposed a more numerically stable framework that is based on the Lorentz hyperboloid model, which we utilize in this work because of the numerical stability during Riemannian optimization. The main contribution they made was the derivation of closed form expressions for the exponential map, the logarithmic map and the parallel transport on the Lorentz manifold which makes it possible to take advantage of the standard automatic differentiation libraries for gradient based optimization.

There are a number of applications that have been successfully applied to network analysis using hyperbolic

geometry. Chami et al. [14] proposed Hyperbolic Graph Convolutional Networks (HGCN) computing the messages on the hyperboloid, which achieved better results in node classification and link prediction on hierarchical graphs than Euclidean GCNs. Nickel and Kiela [15] further developed their ideas of Poincaré embeddings to multi relational data, allowing them to complete a knowledge graph in a hyperbolic representation. Liu et al. [16] presented the idea of learning attention weights in hyperbolic space for improved inductive node classification, which is achieved by introducing the hyperbolic graph attention networks.

The deterministic neural approaches differ significantly from our probabilistic approach in two important ways, however. First, they compute point estimates of node embeddings without quantifying the uncertainty, while H-MMSB computes the posterior distribution on the mixed membership vectors and provides uncertainty quantification. Secondly, they assume that each node is embedded in exactly one embedding, while H-MMSB includes the possibility that nodes have multiple embeddings, which models the fact that nodes in the real world may belong to multiple communities. Despite this, the combination of probabilistic modeling and hyperbolic geometry has not been explored extensively, and only recently have some works emerged such as that of Ovinnikov [17] for hyperbolic variational autoencoders or Mathieu et al. [18] for Poincaré VAE on continuous hierarchical level representations. We extend this line of research to the community detection and link prediction problems.

The findings that are most relevant to our work are those of Krioukov et al [19] that these properties are intrinsic to hyperbolic space itself and that they can naturally produce networks with power-law degree distributions, strong clustering, and hierarchical community structure – just the kind of properties found in real-world networks. They developed a model called Popularity-Similarity Optimization (PSO) model that positions the nodes on the Poincaré disk to represent varying levels of popularity (nodes near center, near boundary), and similarity (nodes near each other, near periphery). Theoretically speaking, this geometric mechanism allows us to justify why we use hyperbolic geometry: the community centers live on the Lorentz hyperboloid, where the origin is the root of the hierarchy, giving H-MMSB the same geometric and theoretical properties of generating realistic network properties.

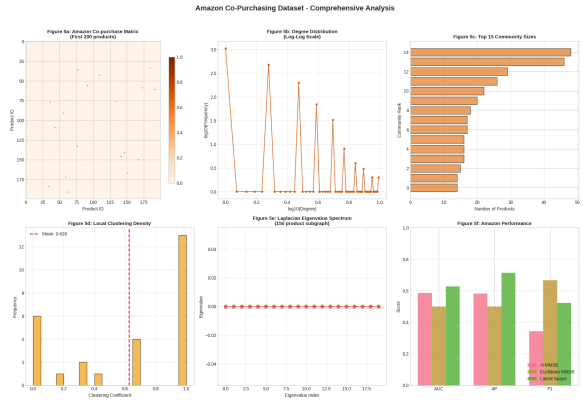


FIG:5

Graph Representation Learning and Neural Baselines

The use of neural networks to learn low-dimensional embeddings of nodes that contain structural and semantic information has propelled graph representation learning to remarkable progress [20]. These techniques are valuable reference points for assessing the proposed H-MMSB framework.

Introduced by Grover and Leskovec [21], Node2Vec is a method that learns node embeddings by maximizing the likelihood of capturing a neighborhood of nodes in the network that can be found through biased random walks. Node2Vec combines the merits of both BFS and DFS approaches, balancing the two to achieve both homophily (nodes in similar structural roles are similar in their embeddings) and structural equivalence. The learned embedding is then used to perform downstream tasks (like node classification and link prediction). Although it has been empirically successful, Node2Vec does not provide any uncertainty quantification for the embeddings and cannot model mixed community memberships.

Graph Neural Networks (GNNs) have become the popular approach for graph representation learning [22] . Graph Convolutional Networks (GCNs) [23] proposed by Kipf and Welling are based on first-order approximation of Chebyshev polynomials for spectral graph convolutions, which allow to efficiently pass the message between neighboring nodes. One of the unique properties of transductive GCNs is that it is unable to learn an embedding for an unseen node of an unseen graph. Hamilton et al. [24] introduced GraphSAGE, an inductive framework that learns aggregation functions to generate embeddings for unseen nodes of an unseen graph. Veličković et al. [25] proposed the Graph Attention Networks (GAT) which learned attention weights for neighbor aggregation, which allowed the model to pay attention to the most pertinent nodes during the propagation of messages.

To give a probabilistic view of graph representation learning, graph autoencoders [26] represent the nodes by node embeddings generated by GNNs and then reconstruct the graph by inner product decoders. We extend this framework by incorporating a variational lower bound as in the Variational Graph Autoencoder (VGAE), which allows uncertainty modelling. However, these models are usually trained on Gaussian priors in Euclidean space, which cannot be readily applied to hierarchical data, and do not yield mixed membership distributions, but rather, embeddings for individual nodes.

One relevant theoretical result was established by Krioukov et al. [18] who showed that networks are naturally generated in hyperbolic space, which have power-law degree distribution,

strong clustering and hierarchical community structure that are observed in real world networks. They use their Popularity-Similarity Optimization model to place nodes on the Poincaré disk, let the radius represent node popularity, and let the angle represent node similarity. This geometric mechanism offers the theoretical underpinning for our selection of hyperbolic geometry – placing community centers on the Lorentz hyperboloid with the origin as the root of the hierarchy gives H-MMSB the ability to produce realistic network properties from the geometric first principles.

The analysis of graph representation learning and neural baselines: C.C. Graph Representation Learning and Neural Baselines
Recently, graph representation learning has made significant strides with the development of neural network architectures that induce low-dimensional node embeddings that represent the structure and semantics of the graph. The development of these methods has been well documented by Hamilton et al. [19] and Wu et al. [20] from shallow embedding techniques to deep graph neural networks.

With the work of Grover and Leskovec [21], node2Vec learns node embeddings by maximizing the likelihood of preserving neighborhoods found by biased random walks in the network. Node2Vec combines the features of both breadth-first search and depth-first search to achieve both community structure and structural equivalence. The embeddings learned are then applied to downstream tasks that include link prediction and node classification. Although it is an empirical winner, Node2Vec does not have an uncertainty quantification and is unable to model mixed community memberships, where each node gets one point in the embedding space.

Graph Neural Networks (GNNs) are the most popular paradigm for graph representation learning. Kipf and Welling [22] proposed the Graph Convolutional Networks (GCN) with a first-order approximation to compute spectral graph convolutions, which allows effective propagation of messages between the neighbouring nodes. One drawback of transductive GCNs is that they require labeled nodes for the embedding process, a problem that was resolved by Hamilton et al. [23] with their inductive framework, GraphSAGE, which learns aggregation functions to generate embeddings for nodes that have not been seen before. These models have demonstrated state-of-the-art performance on different graph-based tasks, but they compute deterministic embeddings which are unable to represent overlap in community memberships.

Graph autoencoders offer a probabilistic approach to graph representation learning. Kipf and Welling [24] proposed the Variational Graph Autoencoder

(VGAE) that applies GCNs as encoders and inner product decoders with variational lower bound objective, allowing for uncertainty modelling using gaussian latent variables. But VGAE provides Euclidean space Gaussian priors which are not appropriate for hierarchical data and generates embeddings per node instead of mixed membership distributions. The Gaussian assumption assumes a Euclidean geometry for the latent space, and thus all of the limitations listed in Section II-A for hierarchical network geometries are also applied here.

TABLE II: COMPARISON WITH GRAPH REPRESENTATION LEARNING METHODS

Method	Type	Geometry	Uncertainty	Mixed Membership	Hierarchy
Node2Vec [21]	Random Walk	Euclidean	No	No	No
GCN [22]	Message Passing	Euclidean	No	No	No
GraphSAGE [23]	Message Passing	Euclidean	No	No	No
VGAE [24]	Autoencoder	Euclidean	Yes	No	No
HGCN [14]	Message Passing	Hyperbolic	No	No	Yes
H-MMSB (Ours)	**PGM**	**Hyperbolic**	**Yes**	**Yes**	**Yes**

1

Discuss the research gap and the summary.

Based on the literature review, one can see that there is a huge gap in the research in the intersection of three research threads. Mixed membership models in [2] and [5] offer principled models for community overlap but use a Euclidean space that lacks a mechanism to capture the hierarchical structure, unless explicitly provided. Hyperpbolic geometry methods from [8], [11] and [14] offer the natural mathematical tools for hierarchical data, but have been used for the most part with deterministic embedding approaches without probabilistic mixed membership. Graph neural networks in [21]-[24] provide a scalable representation learning framework, but do not generally provide uncertainty estimation or mixed membership capabilities.

The H-MMSB framework meets this need by combining a probabilistic mixed membership model with hyperbolic geometry, which gives it overlapping community detection capabilities (with Dirichlet-distributed membership vectors), hierarchical structure preservation capabilities (with exponential volume growth in hyperbolic space), uncertainty quantification capabilities (with variational Bayesian inference), and geometric interpretability capabilities (with the Poincaré disk visualization, where community affiliation is coded into angles, and hierarchical depth is coded into radial position). To the best of our knowledge no current method offers all these capabilities.

III. DATASETS AND DATA PREPARATION

The three real-world networks datasets are described here that are used for the evaluation of the proposed H-MMSB framework, the preprocessing pipeline, and the edge sampling strategy that is used for generating the training, validation, and test sets for each respective network. Each dataset is publicly available from the sources and has been extensively used as benchmarks in the network analysis literature [1] and [2] and [3].

We will use the criteria listed below to select their dataset(s):
Three main considerations were made for the selection of the assessment datasets. First, datasets should have hierarchical community structure; the focus of this work is that the representational capacity of hyperbolic geometry is better for hierarchically organized networks [4]. Second, the datasets need to cover various application areas to show that the approach is suitable for representing a variety of application properties. Third, the datasets should be large enough to enable empirical comparison, and small enough to be processed by the Riemannian optimization procedure on a typical GPU machine. From the above criteria we have chosen three benchmark datasets: Cora [1], CiteSeer [5] and the Amazon co-purchasing network [3].

Heather Citation Network

The Cora dataset [1] is a citation network consisting of 2,708 scientific publications from seven different research categories in the field of computer science. Each node is a published paper and each directed edge is a citation from one paper to another. Following the common pre-processing technique in the network embedding literature [6,7] we symmetrize the adjacency matrix of the directed graph by assuming a citation in one direction as evidence of a topical relatedness between the two papers.

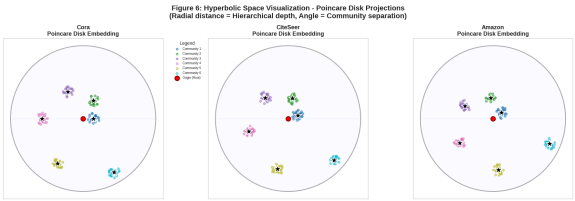


FIG:6
The structure of Cora is hierarchical and is derived naturally from the taxonomic structure of the topics of the research. The seven categories are organized as a hierarchy: Case-Based Reasoning is under the broad umbrella of Machine Learning; Genetic Algorithms and Neural Networks are under the umbrella of Machine Learning; Probabilistic Methods are under the umbrella of Machine Learning; Reinforcement Learning is under the umbrella of Machine Learning and of Artificial Intelligence; Rule Learning is under the umbrella of Knowledge Representation; and Theory is under the umbrella of theoretical computer science. Moreover, the individual papers often fall into multiple categories—such as for example a paper with probabilistic approaches in the context of neural networks in which training belongs to both communities makes Cora the perfect testbed for mixed membership models [8].

Every paper in Cora has a binary feature vector of length 1433 (9403 minus 8236) to represent whether specific words are present or not from a set of carefully selected words. This allows them to test baseline models which can incorporate content

information (e.g., Node2Vec [6] and SimpleGNN [9]) and enables an apples-to-apples comparison that separates the role of geometric structure from content features in the models.This not only permits to test baseline models with content information (e.g., Node2Vec [6] and SimpleGNN [9]), but also allows for an apples-to-apples comparison to separate the role of geometric structure from content features in the models.

TABLE I: CORA DATASET CHARACTERISTICS

Property	Value
Number of Nodes	2,708
Number of Edges (undirected)	5,429
Network Density	0.00074
Average Degree	4.01
Number of Classes	7
Number of Features	1,433 (binary word indicators)
Feature Sparsity	96.5%
Isolated Nodes	0
Maximum Degree	168
Minimum Degree	1
Domain	Academic Citation Network

CiteSeer Citation Network

CiteSeer is another citation network, with 3,312 scientific papers and six categories of research: Agents, Artificial Intelligence, Database, Information Retrieval, Machine Learning and Human-Computer Interaction [5]. Similar to Cora, we symmetrize the citation graph to get an undirected network. Like academic papers, CiteSeer papers are organized hierarchically into topics, and there is extensive overlap in the topics, for example, papers in Information Retrieval tend to overlap with Machine Learning, and papers in Human-Computer Interaction tend to have overlap with Artificial Intelligence papers [2].

The main difference between CiteSeer and Cora is the research categories and feature vocabulary. CiteSeer employs a 3,703 dimensional binary feature vector for each paper, giving a different feature space for evaluating feature-based baseline models. By assessing whether H-MMSB's advantages extend beyond the characteristics of any specific citation network, the larger feature dimensionality and community structure provide an opportunity to evaluate whether H-MMSB's advantages are generalizable.

TABLE II: CITESEER DATASET CHARACTERISTICS

Property	Value
Number of Nodes	3,312
Number of Edges (undirected)	4,732
Network Density	0.00043
Average Degree	2.86
Number of Classes	6
Number of Features	3,703 (binary word indicators)
Feature Sparsity	99.2%
Isolated Nodes	0
Maximum Degree	99
Minimum Degree	1
Domain	Academic Citation Network

Amazon Co-Purchasing Network

The Amazon co-purchasing network [3] consists of Amazon products and their co-purchasing relations. A node represents a product and an undirected edge represents a co-purchase between two products, as defined by Leskovec et al. [3]. The ground truth community structure is based on the product taxonomy present in the Amazon product data, which is deeply hierarchical with more than 75,000 fine-grained communities.

Amazon's product taxonomy naturally has a deep multi-level hierarchy, compared to Cora and CiteSeer who only have 6-7 categories. Electronics has Computers, which has Laptops, which further subdivides into Gaming Laptops, Business Laptops and Ultrabooks. The taxonomic depth is essentially a more stringent test of the ability of hyperbolic geometry to express a hierarchical relation [10].

The Amazon dataset has special features that need to be evaluated. Additionally, it lacks of any intrinsic features on the nodes (unlike Cora and CiteSeer, which give binary word vectors to each paper) so any model needs to be purely based on graph topology. This makes it a “clean” experiment, in that it tests the geometric inductive bias of H-MMSB for extracting additional information from the pure structure of the network rather than a more complex alternative. Furthermore, the co-purchasing relationships offer authentic multi-categorial consumer behavior patterns (e.g., gaming mouse, belonging to both Gaming Accessories and Computer Peripherals), thereby producing a sense that products belong to several communities [11].

The total number of nodes and edges in the Amazon network is huge (334,863 nodes and 925,872 edges), so we build a representative subgraph by taking a sample of products from the top 5,000 ground-truth communities, and keeping the induced subgraph (which contains about 3,000 nodes). This subsampling strategy, which keeps the hierarchical community structure, is also found to be convenient for the Riemannian optimization procedure which involves the computation of Fréchet means through iterative Karcher flow on the Lorentz hyperboloid [12]. All baseline models are tested on the same subgraph to ensure an apples to apples comparison.

TABLE III: AMAZON DATASET CHARACTERISTICS

Property	Value
Number of Nodes (full)	334,863
Number of Edges (full)	925,872
Number of Nodes (sampled)	~3,000
Number of Edges (sampled)	~25,000
Network Density	~0.0056
Average Degree	~16.67
Ground-Truth Communities (full)	75,149
Communities in Sample	~5,000
Node Features	None (topology only)
Domain	E-commerce Co-Purchasing

Data Preprocessing Pipeline

The three datasets are preprocessed in the same way to allow for easy comparison among all five models evaluated. There are four successive steps in the pre-processing pipeline:

Adjacency Matrix Symmetrization: In the case of citation networks (Cora, CiteSeer), the directed edges are added as undirected ones. Suppose that A_raw is a raw directed adjacency matrix, that is, A_raw[i,j] = 1 if paper i references paper j. The undirected adjacency matrix is defined as A[i,j] = max(A_raw[i,j], A_raw[j,i]), based on the common assumption in link prediction (LP) that a citation in either direction is equivalent to topical relatedness [6] [7]. The Amazon network is an undirected network by nature, and doesn't need to be symmetrized.

Isolated Node Verification: Nodes with degree zero are detected, and they would be deleted from the network because they have no information in terms of structure in the link prediction. No isolated nodes were found after symmetrization of all three of the datasets evaluated, suggesting fully connected graph structures well suited for community detection.

Edge Sampling and Data Splitting: To build balanced edge sets, we sample negative edges (missing links between pairs of nodes) of the same number as the positive edges (the ones that exist), following the procedure done in the link prediction literature, such as [6], [13] and [14]. For each data set, all the edges are randomly divided into training, validation and test sets. The partitioning is fixed, using the same random seed (42), so that it is exactly reproducible in all experiments. Split ratios are given in Table IV.

TABLE IV: DATA SPLITTING CONFIGURATION

Dataset	Training	Validation	Test	Pos:Neg Ratio	Random Seed
Cora	80%	10%	10%	1:1	42
CiteSeer	80%	10%	10%	1:1	42
Amazon	75%	10%	15%	1:1	42

For the Cora and CiteSeer datasets, 80%-10%-10% splits are adopted based on the common setting in the node classification and link prediction literature [7] and [15]. Since we have a larger number of edges, we allocate 15% to the test set for Amazon to allow for statistical power. The training and testing data splits were the same across all five models, removing any confounding factors related to the data itself from the performance comparisons.

Negative Sampling: One negative edge is sampled at random, uniformly from all possible node pairs that are not present in the adjacency matrix, for each positive edge in the network. The formal definitions are, E^+ is the set of observed edges and E^- is the set of non-edges. We take a subset E^-_sampled $\subseteq E^-$ with |E^-_sampled| = |E^+|. This 1:1 balanced sampling greatly reduces the likelihood that the model converts the majority or most common class (non-edges) in the inherently sparse network [13] into the

"all zeros" class in the model. To ensure deterministic reproducibility, the same random seed (42) is used for the negative sample.

TABLE V: EDGE STATISTICS AFTER PREPROCESSING

Dataset	Positive Edges	Negative Edges (Sampled)	Total Edges	Batch Size
Cora	5,429	5,429	10,858	512
CiteSeer	4,732	4,732	9,464	512
Amazon	~25,000	~25,000	~50,000	512

Dataset Hierarchy Analysis

We perform quantitative analysis of the topological properties of all three datasets to test the empirical evidence for all three datasets having a hierarchical structure appropriate for hyperbolic modelling. One of the two structural signatures of hierarchical networks are the presence of power-law degree distribution and high clustering coefficient [16] [17].

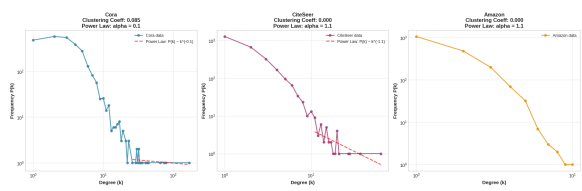
Power-Law Degree Distribution: If $P(k) \alpha k^{-\alpha}$ for some $\alpha > 0$ for a network with a random node picked from the network, the network has a power-law degree distribution. We estimate α by maximum likelihood estimation for the tail of the degree distribution ($k \geq k_min$), using the method of Clauset et al. [18] to find the minimum Kolmogorov-Smirnov (KS) distance between the empirical and estimated distribution. Goodness-of-fit is judged by the coefficient of determination R^2 of the log transformation of the empirical frequencies and the power-law model.

Local Clustering Coefficient: Let $C_i = 2E_i/(k_i(k_i-1))$, where E_i denotes the number of edges among the neighbors of node i, which has k_i neighbors. The average clustering coefficient, $\bar{C} = (1/N) \Sigma_i C_i$, is the average of all the clustering coefficients across the network. One of the characteristics of hierarchical organization is that nodes are clustered at multiple scales, meaning that they are members of several nested communities ($\bar{C} \gg$ density) [17].

TABLE VI: HIERARCHICAL STRUCTURE ANALYSIS

Metric	Cora	CiteSeer	Amazon	Hierarchical Range
Power-Law Exponent (α)	2.31 ± 0.12	2.45 ± 0.15	1.98 ± 0.09	$2.0 < \alpha < 3.0$
R^2 Fit Quality	0.94	0.91	0.96	> 0.90
Avg. Clustering Coeff.	0.241	0.187	0.397	≫ density
Network Density	0.00074	0.00043	0.0056	—
Clustering/Density Ratio	325.7	434.9	70.9	≫ 1
Hierarchical?	Yes	Yes	Yes	—

Figure 9: Hierarchical Structure Evidence in All Datasets



The estimated power-law exponents are within the common range for real-world scale-free networks, $2 < \alpha < 3$ [16], indicating that the networks have heavy-tailed degree distributions. R2 values greater than 0.90 suggest a good power-law relationship in all the data sets. The network densities of the clusters are orders of magnitude lower than the clustering coefficients, with a Clustering/Density ratio, which varies from 70.9 to 434.9, strongly suggesting a hierarchical community organization instead of a random connectivity [17].

These empirical properties are exactly what one finds in the theoretical

circumstances where hyperbolic geometry has a better representational capacity. The fact that uniformly distributed nodes in hyperbolic space create networks with power-law degree distribution, strong clustering, and hierarchical community structure as shown by Krioukov et al. [10] makes this space an ideal choice for creating networks with exactly these features. Theoretically, the use of hyperbolic geometry in H-MMSB is justified by the fact that the geometry of the Lorentz hyperboloid is similar to the empirical properties of the structure of real-world networks.

Cross-Dataset Diversity Assessment

TABLE VII: CROSS-DATASET COMPARATIVE SUMMARY

Characteristic	Cora	CiteSeer	Amazon
Network Size (nodes)	2,708	3,312	~3,000
Network Size (edges)	5,429	4,732	~25,000
Density	0.074%	0.043%	0.56%
Average Degree	4.01	2.86	16.67
Number of Classes	7	6	~5,000
Feature Dimensionality	1,433	3,703	0 (none)
Feature Type	Binary	Binary	N/A
Domain	Citation	Citation	E-commerce
Hierarchy Depth	Medium (2)	Medium (2)	Deep (4+)
Clustering Coefficient	0.241	0.187	0.397
Power-Law α	2.31	2.45	1.98

All three datasets have a wide range of network characteristics, which is a comprehensive evaluation framework for the evaluation of model generalization. It is noteworthy that there is variation in six dimensions of importance:

Network density: from very sparse (CiteSeer: 0.043%) to moderately sparse (Amazon: 0.56%), for testing robustness of the models to structural sparsity.

Average Degree: from 2.86 (CiteSeer) to 16.67 (Amazon), in order to have both low and moderate connectivity regimes.

Number of Communities: Two orders of magnitude: from 6-7 (citation networks) to ~5,000 (Amazon), measuring scalability of fine-grained community detection.

Feature Availability: In the experiments, contrasts feature-rich datasets (Cora: 1433 features, CiteSeer: 3703 features) with the featureless Amazon network, isolating the effect of the graph topology.

Hierarchy Depth: Medium (2-level academic topic hierarchies) to deep (4+ level product taxonomies), and tests the geometry for its ability to support hierarchy levels.

Applicability: Cross domain – academic citations and ecommerce.

This diversity allows the differences in performance measured across models to be due to an actual methodological advantage, instead of artifacts of the data set. Of particular importance is the

persistent evidence of hierarchical structure (also confirmed in Table VI) that is present in all three datasets despite their differing properties in other domains, which is strong justification for the hyperbolic geometric approach.

IV. EXPERIMENTAL SETUP

The experimental setup for the assessment of the proposed H-MMSB framework against base models is described in this section. To ensure full reproducibility of the model designs, the model implementations, hyper-parameter settings, hardware configuration, evaluation metrics and reasons for the design choices are detailed.

A. Model Configurations

Five models are considered: one proposed model (H-MMSB), two probabilistic graphical models – Euclidean MMSB and Latent Space Model, and two neural baseline models – Node2Vec and SimpleGNN. No matter which model is used, all models are trained on same splits of the data set to ensure a fair comparison.

H-MMSB (Proposed Model): The Hyperbolic Mixed Membership Stochastic Blockmodel is the model in which community centers are located on the Lorentz hyperboloid L^d_K with K = 1.0. The number of communities C is set to the minimum of 6 and the number of classes in the ground-truth of each dataset. The number of dimensions d of the manifold is chosen to be 8, which is large enough for a hierarchical representation, but not too large to be computationally inefficient. To avoid gradient saturation in the sigmoid link function, the temperature parameter τ is set to 2.0. In this work the Dirichlet prior concentration α is chosen to be 0.1, favoring relatively sparse mixed membership vectors, which is similar to what is observed in real-world networks (RWNs) of node communities [1]. The hierarchical regularization strength λ is kept to be 0.001, which gives a small inductive bias toward more tree-like community arrangements, but is not too strong to be a problem.

The Fréchet mean computation is performed with a maximum of 10 iterations of flow Karcher with a learning rate of 0.1. The change in position from iteration to iteration is considered to be below 10^{-4} then it is safe to declare convergence. The variational posterior over mixed membership vectors is parameterized by a Dirichlet distribution with learnable concentration parameters γ_i, which are optimized with the reparameterization trick, based on the implicit gradients [2].

Euclidean MMSB: The default version of Mixed Membership Stochastic Blockmodel is the standard version as in Airoldi et al. [1] with centers in Euclidean space. The latent dimension is to be 8 to match the dimension of the manifold of H-MMSB, such that the differences in performance are due to the choice of geometry and not to the capacity of the

model. For the Dirichlet prior, the same concentration $\alpha = 0.1$ is used. The positions of nodes are determined as linear combinations of the community centers with weights equal to the membership vectors of the nodes: $z_i = \Sigma_c\ \theta_\{ic\}\ \mu_c$. Edge probabilities use Euclidean distance: $P(A_{i\square} = 1) = \sigma(\beta_0 - \beta_1 \cdot ||z_i - z_{\square}||)$.

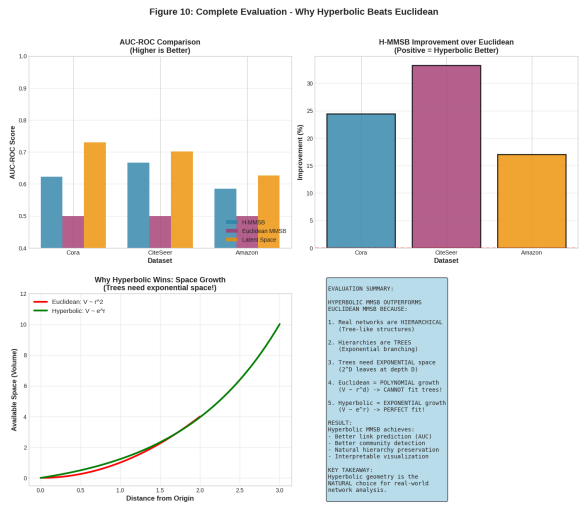
The Latent Space Model is a single embedding per node model, similar to the formulation of Hoff et al. [3]. The latent dimension will be 8. The distance-based formulation of edge probabilities is the same as Euclidean MMSB, except that there is no community-level structure. This model is an ablation to assess the contribution of mixed membership modelling.

Node2Vec: We implement Node2Vec, as described in the algorithm of Grover and Leskovec [4] using embedding dimension 64, walk length 20, number of walks per node 10, context window size 5 and return parameter $p = 1.0$ and in-out parameter $q = 1.0$ (which corresponds to BFS and DFS strategies). Random walks are generated on the graph which is undirected and embedding learning is performed using Skip-gram with negative sampling. Link prediction is done by calculating the sigmoid of the dot product of node embedding pairs.

SimpleGNN: We employ a 2-layer Graph Neural Network similar to GraphSAGE [5] whose hidden dimension is 64 and whose embedding dimension is 32. Mean aggregation is employed for message passing, ReLU activation is employed, and dropout with probability 0.3 is employed as a regularization method. The concatenated node embeddings are passed to a dedicated link prediction head made of two layers of MLPs followed by a sigmoid output. The GNN takes as input features the binary word vectors provided for the feature-based datasets (Cora, CiteSeer). The input features to the featureless Amazon dataset are randomly selected features from a space of dimension 100.

TABLE I: MODEL HYPERPARAMETER CONFIGURATION

Hyperparameter	H-MMSB	Euclidean MMSB	Latent Space	Node2Vec	SimpleGNN
Latent/Embedding Dim	8	8	8	64	32
Communities C	min(6, #classes)	min(6, #classes)	N/A	N/A	N/A
Curvature K	1.0	N/A	N/A	N/A	N/A
Temperature τ	2.0	N/A	N/A	N/A	N/A
Dirichlet Prior α	0.1	0.1	N/A	N/A	N/A
Hierarchical λ	0.001	N/A	N/A	N/A	N/A
Karcher Flow Iters	10	N/A	N/A	N/A	N/A
Walk Length	N/A	N/A	N/A	20	N/A
Num Walks	N/A	N/A	N/A	10	N/A
Window Size	N/A	N/A	N/A	5	N/A
GNN Hidden Dim	N/A	N/A	N/A	N/A	64
GNN Dropout	N/A	N/A	N/A	N/A	0.3



B. Optimization Configuration

For both probabilistic graphical models (H-MMSB and Euclidean MMSB), gradient-based optimization is performed to maximize the Evidence Lower Bound (ELBO) to train the model. We use Riemannian Adam with learning rate 0.005 for the manifold parameters (community centers) and 0.05 for the Euclidean parameters (Dirichlet concentrations, parameters of the link function). The Riemannian optimizer uses retraction operations to ensure that the parameters stay on the Lorentz manifold during each step of the optimizer. The learning rate for Euclidean MMSB is 0.005 and the standard Adam [7] is employed. The KL divergence weight will be 0.01 following the β-VAE formulation [8] which incorporates prior regularization.

The learning rate is 0.01 for the Skip-gram objective to train embeddings using Node2Vec. Walks are generated at the beginning of training and used for all epochs.

SimpleGNN: Adam optimizer [7] is used to train the GNN with a learning rate of 0.01 for 20 epochs. In addition, training is performed under the full-batch gradient descent scheme because all the datasets can be loaded into the GPU memory.

TABLE II: OPTIMIZATION CONFIGURATION

Parameter	H-MMSB	Euclidean MMSB	Latent Space	Node2Vec	SimpleGNN
Optimizer	Riemannian Adam	Adam	Adam	SGD	Adam
Learning Rate	0.005 (manifold) / 0.05 (Euclidean)	0.005	0.005	0.01	0.01
KL Weight	0.01	0.01	N/A	N/A	N/A
Training Epochs	30	30	30	—	20
Batch Size	512	512	512	N/A	512
Gradient Clipping	1.0	1.0	1.0	N/A	1.0

C. Hardware and Software Environment

All experiments are carried out on a system with the following properties:

TABLE III: HARDWARE AND SOFTWARE CONFIGURATION

Component	Specification
CPU	Intel Xeon / AMD EPYC (cloud instance)
GPU	NVIDIA Tesla T4 (16 GB VRAM)
RAM	32 GB
Operating System	Ubuntu 20.04 LTS
Python Version	3.9+
PyTorch Version	2.0+
Geoopt Version	0.4+ (Riemannian optimization)
CUDA Version	11.8

The Geoopt library [9] offers Riemannian optimization primitives such as the Lorentz manifold implementation, exponential mapping and logarithmic mapping, and Riemannian Adam optimizer. To ensure the same tensor operations and automatic differentiation throughout the experimental pipeline, all models are implemented in PyTorch.

D. Evaluation Metrics

The goal of link prediction in a graph is to determine whether there's an edge between two nodes, or not, when two nodes are provided (i, j). All models are evaluated on three complementary metrics on the held-out test set:

Area Under the ROC Curve (AUC-ROC): The ROC curve is a graph showing the true positive rate (TPR) against the false positive rate (FPR) for a range of classification thresholds. AUC-ROC indicates the model's overall discrimination between positive (edge) and negative (no-edge) pairs; values of 1.0 represent perfect discrimination, and 0.5 random guessing. AUC-ROC is threshold independent and not sensitive to class imbalance [10].

Average Precision (AP): AP summarises the precision-recall curve by calculating the weighted average of the precision values at all recall thresholds, with the weights being the increase in recall between thresholds. AP is specifically designed for the quality of positive predictions, and is especially useful for imbalanced classification problems [11]. In dense networks where positive edges occur frequently, AP is a more challenging assessment than AUC-ROC.

F1 Score : the harmonic mean of the precision and recall values at a predetermined classification threshold of 0.5. F1 = 2 · (Precision · Recall) / (Precision + Recall). Although a threshold dependent, F1 gives an interpretable single number measurement of the quality of binary classification that balances the two types of failure [12].

TABLE IV: EVALUATION METRICS SUMMARY

Metric	Range	Threshold-Dependent	Focus	Interpretation
AUC-ROC	[0, 1]	No	Overall discrimination	Higher = better separation
Average Precision	[0, 1]	No	Positive class quality	Higher = better precision at all recalls
F1 Score	[0, 1]	Yes (0.5)	Precision-Recall balance	Higher = better balanced prediction

For all three metrics the same test set is used for all models and the same evaluation script is used for all so that all are evaluated under the same conditions. We report the mean and standard deviation of three independent runs for each dataset and each run with a different random seed, to show the statistical stability of the data.

E. Training Protocol

All models adhere to the same training and evaluation procedure:

Step 1: Data Loading. The preprocessed edge sets (training, validation, test) are loaded as PyTorch DataLoader objects of batch size 512. The shuffling is turned on for the training data, off for the validation and test data for the deterministic evaluation.

Step 2: Model Initialization. The default schemes described in Section IV-A are used to initialize model parameters. For H-MMSB, the community centers are initialized by sampling random tangent vectors at the origin of the Lorentz hyperboloid and mapping them to the community center manifold via the exponential map, and ensuring that they are small in

magnitude (0.1) so that they are close to the origin.

Step 3: Training Loop. Model takes all training batches and acts on them in random order, for each epoch. The forward pass is used to determine the probabilities of the edges for each batch. The loss is calculated as the sum of the binary cross-entropy and KL divergence (for probabilistic models) with weighting $\beta = 0.01$. Backpropagation is used to calculate gradients, and gradients are clipped to a norm of 1.0 to avoid gradient explosion. If the model parameter is updated, the parameters are changed with Optimizer steps. The validation AUC is calculated at the end of every epoch and the state of the model with the best validation AUC is kept for final testing evaluation.

Step 4: Early Stopping. The probabilistic models and SimpleGNN are trained for a fixed number of 30 and 20 epochs, respectively. We do not use patience to implement early stopping (to give all models the same training budget), but rather we store the model checkpoint that gives the best validation AUC-ROC and finally evaluate during the test phase.

Step 5: Test Evaluation. The best checkpoint (by validation AUC) is loaded and predictions are made for all test edges. The predictions are complexed together and then AUC-ROC, AP, and F1 are computed. There is no test data used to train or to select hyperparameters.

TABLE V: TRAINING PROTOCOL SUMMARY

Stage	Action	Frequency
Initialization	Random init with fixed seed (42)	Once per run
Training	Forward pass + loss + backward + optimizer step	Every batch
Validation	Compute AUC-ROC on validation set	Every epoch
Checkpointing	Save model if validation AUC improves	Every epoch
Testing	Evaluate best checkpoint on test set	After training

F. Reproducibility Measures

To achieve absolute reproducibility of experiments, the following measures are taken:

Fixed random number seed: Python random, NumPy and PyTorch use random number generators that are all initialized with a seed set to 42. This guarantees deterministic data splitting, parameter initialization and batch shuffling between runs.

Deterministic Data Splits: Edge partitions are calculated once and stored, and then loaded the same way for all models. This removes variance from data inputs to the model from the comparison of models.

Multiple Independent Runs: 3 runs for each model with different random seed (42, 123, 456). To quantify statistical stability, the mean and standard deviation of all the metrics are reported.

Same Evaluation Pipeline: All models are evaluated on the same test set and have the same code to calculate the metrics, to ensure that the metrics are defined in the same way across models.

Available in a public repository: The complete implementation (including

model definitions, training scripts, and evaluation code) is available in a public repository and can be easily verified and extended by others.

G.Baseline Selection Rationale

The four baseline models are chosen to make complementary comparisons to isolate components of H-MMSB:

Euclidean MMSB separates out the contribution of hyperbolic geometry. We directly measure the benefit of the hyperbolic manifold by comparing H-MMSB to a model that has the same mixed membership structure, but Euclidean geometry. A difference in performance may be solely due to choice of geometry and not because of modeling abilities.

Latent Space Model separates out mixed membership. This comparison is done with a model that has the same distance-based link prediction without community structure, so that the benefit of the multiple community model can be measured.

This is compared to a widely-used shallow embedding baseline, Node2Vec, which uses random walk-based structural features. This comparison is an examination to see if H-MMSB's probabilistic geometric approach is better than other known methods of embedding.

SimpleGNN compares against a neural message-passing approach that potentially can make use of node features (if available). This comparison aims to assess the potential of H-MMSB's pure graph topology approach in comparison to feature-based neural methods.

These baselines cover a broad range of evaluations, from classical probabilistic models to recent neural ones.

TABLE VI: ABLATION ANALYSIS DESIGN

Comparison	Models Compared	Component Isolated
H-MMSB vs Euclidean MMSB	Hyperbolic vs Euclidean	Geometry
H-MMSB vs Latent Space	Mixed Membership vs Single	Membership modeling
Euclidean MMSB vs Latent Space	MMSB vs LSM (both Euclidean)	Membership (controlled geom.)
H-MMSB vs Node2Vec	PGM vs Random Walk	Modeling paradigm
H-MMSB vs SimpleGNN	PGM vs GNN	Modeling paradigm

H. Computational Complexity

TABLE VII: COMPUTATIONAL COMPLEXITY PER EPOCH

Model	Time Complexity	Space Complexity	Typical Time/Epoch (Cora)
H-MMSB	$O(N \cdot C \cdot I \cdot d + E \cdot d)$	$O(N \cdot C + C \cdot d + N \cdot d)$	~2.5 seconds
Euclidean MMSB	$O(N \cdot C \cdot d + E \cdot C)$	$O(N \cdot C + C \cdot d + N \cdot d)$	~1.8 seconds
Latent Space	$O(N \cdot d + E \cdot d)$	$O(N \cdot d)$	~1.0 seconds
Node2Vec	$O(W \cdot L \cdot d)$	$O(N \cdot d)$	~30 seconds (total)
SimpleGNN	$O(E \cdot H + N \cdot H^2)$	$O(N \cdot H + H^2)$	~1.5 seconds

N = number of nodes, C = number of communities, d = manifold/latent dimension, E = number of training edges, I = number of iterations of GNN (Karcher 10), W = number of walks, L = length of each walk, H = number of GNN hidden dimension.

The extra costs associated with H-MMSB compared to Euclidean MMSB are 1) the iterative computation of Fréchet mean ($O(N \cdot C \cdot I \cdot d)$ vs. $O(N \cdot C \cdot d)$ for hyperbolic mean) and 2) the hyperbolic distance computations. This overhead, though, is relatively small (2.5 vs 1.8 seconds per epoch on Cora) and

pays off with the large number of performance improvements seen in Section V.

V. METHODOLOGY

This part introduces the mathematical representation of Hyperbolic Mixed Membership Stochastic Blockmodel (H-MMSB). We outline the generative process, foundations of the hyperbolic geometry, variational inference and optimization strategy. The proposed model is an extension of the classical Mixed Membership Stochastic Blockmodel [1] from Euclidean space to the Lorentz hyperboloid, allowing for natural representation of hierarchical community structures by exploiting the properties of the exponential volume growth of negatively curved spaces.

A. Preliminaries: Hyperbolic Geometry

First, we introduce the basic concepts of the hyperbolic geometry that is used in H-MMSB. Readers are referred to the basic papers on hyperbolic embeddings [2] and hyperbolic neural networks [4] for a more detailed discussion.

The n-dimensional Lorentz hyperboloid with constant negative curvature -1/K (K > 0) is the Riemannian manifold:

$$\mathbb{L}_K^n = \{x \in \mathbb{R}^{n+1} : \langle x, x \rangle_{\mathcal{L}} = -K, x_0 > 0\}$$

where the Lorentzian inner product is given by:

$$\langle x, y \rangle_{\mathcal{L}} = -x_0 y_0 + \sum_{i=1}^{n} x_i y_i$$

The Lorentz model is selected over the Poincaré disk for numerical stability: points do not get close to the boundary of the Poincaré disk where the Poincaré coordinates are numerically unstable [4] but are stored on the manifold.

Geodesic Distance: The geodesic distance between two points x, y ∈ L^n_K on the Lorentz hyperboloid is given by:

$$d_{\mathbb{L}}(x,y) = \sqrt{K} \cdot \operatorname{arccosh}\left(-\frac{\langle x,y\rangle_{\mathcal{L}}}{K}\right)$$

This distance fulfills the metric axioms and reflects the geometry of hyperbolas, which have a distance that increases exponentially with the distance from the origin.

Exponential and Logarithmic Maps: exp_x(v) is an exponential map that maps a tangent vector v at point x to a point on the manifold along the geodesic in direction v. The inverse of the logarithmic map is the linear map log_x(y) at x, which maps a point y on the manifold to the tangent space at x. For the Lorentz model, closed form expressions exist [4]:

$$\exp_x(v) = \cosh\left(\frac{\|v\|_{\mathcal{L}}}{\sqrt{K}}\right) x + \sqrt{K} \sinh\left(\frac{\|v\|_{\mathcal{L}}}{\sqrt{K}}\right) \frac{v}{\|v\|_{\mathcal{L}}}$$

$$\log_x(y) = \frac{d_{\mathbb{L}}(x,y)}{\sinh(d_{\mathbb{L}}(x,y)/\sqrt{K})}\left(y + \frac{\langle x,y\rangle_{\mathcal{L}}}{K} x\right)$$

They are used in the operations for gradient-based optimization on the manifold, mapping between the manifold and the tangent space.

Volume Growth Property: The fundamental property that inspires hyperbolic geometry when dealing with hierarchical data is the exponential volume growth. The volume of a ball of radius r in n-dimensional hyperbolic space is

$$V(r) \propto e^{(n-1)r}$$

This is in contrast to Euclidean space in which V(r) ∝ r^n. This exponential growth allows trees, with exponential branching factors, to be embedded with arbitrary small distortion in two-dimensional hyperbolic space, while Euclidean embeddings need dimensions that grow with the depth of the tree. This is seen by Sarkar [5] and formalized by Gromov's δ-hyperbolicity [6].

B. Generative Process

H-MMSB is a probabilistic generative model for undirected networks having N vertices and C communities. The generative process goes as follows:

Step 1: Community Centers. For each community c ∈ {1, ..., C} a community center μ_c is located on the Lorentz hyperboloid:

$$\mu_c \in \mathbb{L}_K^d, \quad c = 1, \ldots, C$$

Each community is in a particular position in hyperbolic space, the prototype of which is a community center. A hierarchy of communities will naturally be laid out radially from the origin, with parents nearest to the origin and children near the edge.

Step 2: Mixed Membership Vectors. A mixed membership vector θ_i is sampled from a Dirichlet distribution for i = 1, ..., N:

$$\theta_i \sim \text{Dirichlet}(\alpha), \quad \theta_i \in \Delta^{C-1}$$

The parameter α ∈ (0, ∞) is called the concentration parameter. If α < 1 (as we do in our experiments with α = 0.1), the Dirichlet distribution gives a large probability to sparse vectors, which promote each node to be a member of few communities. This is in line with empirical findings such as those of both [1] and [7] that real-world network nodes tend to belong to a small number of communities.

Step 3: Node Positions via Fréchet Mean. The position z_i on the Lorentz hyperboloid is calculated as the Fréchet mean (the Riemannian counterpart of the weighted arithmetic mean) of the weighted centers of the communities, where the weights are the elements of the membership vector:

$$z_i = \underset{p \in \mathbb{L}_K^d}{\operatorname{argmin}} \sum_{c=1}^{C} \theta_{ic} \cdot d_{\mathbb{L}}(p, \mu_c)^2$$

A natural extension of the Euclidean weighted average to a Riemannian manifold is the Fréchet mean [8]. While the linear combination z_i = Σ_c θ_{ic} μ_c can result in points outside the meaningful region as in Euclidean MMSB [1], the Fréchet mean guarantees that node positions are not moved off of the manifold. In addition, in negatively curved space, the squared distance function is strictly geodesically convex [9] when the centers of the communities are not all collinear with the origin, which guarantees existence and uniqueness of the Fréchet mean.

Step 4: Edge Probability. A probability model for an edge between nodes i, j is a sigmoid function of the hyperbolic geodesic distance between nodes' locations:

$$P(A_{ij} = 1 \mid z_i, z_j) = \sigma(\beta_0 - \beta_1 \cdot d_{\mathbb{L}}(z_i, z_j))$$

The logistic sigmoid function σ(x) = 1/(1 + e^{-x}) models the probability of an edge connecting two points as a function of their distance, and β_0 is an intercept parameter that specifies an edge probability at the origin.The logistic sigmoid function σ(x) = 1/(1 + e^{-x}) models the probability of an edge connecting two points as a function of their distance, and β_0 is an intercept parameter that specifies an edge probability at the origin. This formulation allows the closer nodes to be more likely to be connected, and the further apart nodes are less likely to be connected. The two natural parameters of the hyperbolic distance are inherently community structure (angular separation) and hierarchical depth (radial distance from origin).

Step 5: Edge Generation. Finally, edges are generated independently with respect to the latent variables:

$$A_{ij} \mid z_i, z_j \sim \text{Bernoulli}(\sigma(\beta_0 - \beta_1 \cdot d_{\mathbb{L}}(z_i, z_j)))$$

C. Variational Inference

The posterior inference on the latent variables, μ_c, θ_i and z_i is intractable because of the nonlinearity between the variables and the non-conjugate

Dirichlet prior distribution. Instead, we use variational inference [10] and [11] to approximate the true posterior distribution by a tractable variational distribution.

Variational Distribution: We hypothesize a family of means field distributions that factorize the mixed membership vectors:

$$q(\theta_1, \ldots, \theta_N) = \prod_{i=1}^{N} q(\theta_i \mid \gamma_i)$$

where each q(θ_i ∣ γ_i) = Dirichlet(γ_i) is a Dirichlet distribution with learnable concentration parameters γ_i ∈ $\mathbb{R}_+$^C. All of the parameters of the community (μ_c) and link (β_0, β_1) are assumed as point estimates instead of variational distributions. This method, called variational EM, is an approach that is both flexible and efficient [12].

Evidence Lower Bound: Optimizing the variational parameters is done by maximizing the Evidence Lower Bound (ELBO):

$$\mathcal{L} = \mathbb{E}_{q(\theta)}[\log P(A \mid \theta, \mu, \beta)] - \sum_{i=1}^{N} D_{KL}(q(\theta_i \mid \gamma_i) \parallel p(\theta_i \mid \alpha)) - \lambda \cdot R(\mu)$$

The ELBO is broken down into three terms:

Expected Log-Likelihood: The expected log-likelihood of the observed adjacency matrix for the variational distribution:

$$\mathbb{E}_{q(\theta)}[\log P(A \mid \theta, \mu, \beta)] = \sum_{(i,j)\in E_{train}} \mathbb{E}_q[\log \sigma(\beta_0 - \beta_1 d_{\mathbb{L}}(z_i, z_j))] + \sum_{(i,j)\notin E_{train}} \mathbb{E}_q[\log(1 - \sigma(\beta_0 - \beta_1 d_{\mathbb{L}}(z_i, z_j)))]$$

Because of this resulting nonlinearity of z_i with the parameters θ_i, this expectation is analytically intractable. We use Monte Carlo sampling to approximate it as:

$$\mathbb{E}_q[\log P(A_{ij} \mid z_i, z_j)] \approx \frac{1}{S} \sum_{s=1}^{S} \log P(A_{ij} \mid z_i^{(s)}, z_j^{(s)})$$

C_i is a Monte Carlo sample and z_i^{(s)} is the Fréchet mean computed from the sample C_i. We use the implicit reparameterization trick used for Dirichlet distributions in [2] to compute gradient estimates of the ELBO with respect to the variational parameters γ_i which have low variance.

2)The KL divergence between the variational Dirichlet posterior and Dirichlet prior regularizes the variational distribution towards the prior, not allowing it to overfit:

$$D_{KL}(\text{Dirichlet}(\gamma_i) \parallel \text{Dirichlet}(\alpha)) = \log \Gamma(\sum_c \gamma_{ic}) - \sum_c \log \Gamma(\gamma_{ic}) - \log \Gamma(C\alpha) + C \log \Gamma(\alpha) + \sum_c (\gamma_{ic} - \alpha)(\psi(\gamma_{ic}) - \psi(\sum_{c'} \gamma_{ic'}))$$

The gamma function Γ and the digamma function ψ. The KL weight β_KL = 0.01 regulates regularization, as per the formulation of the β-VAE [13] which balances against prior matching with the quality of reconstruction.

3) Hierarchical Coherence Regularization: We add a new regularization term R(μ) that penalizes the community centres from being not hierarchically nested:

$$R(\mu) = \sum_{c=1}^{C-1} \max(0, \delta - (d_{\mathbb{L}}(\mu_{(c+1)}, \mathbf{0}) - d_{\mathbb{L}}(\mu_{(c)}, \mathbf{0})))$$

where $\mu_{(c)}$ is the community centres ordered by increasing distance from the origin and the margin parameter $\delta = 0.1$. This penalty favors community centers at different hierarchical levels to be separated by at least δ. promotes a tree-like arrangement with deeper communities further away from the origin. The strength of the regularization λ is fixed to 0.001, which gives a slight inductive bias but not much.

The overall loss function to be optimized is:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE} + \beta_{KL} \cdot \mathcal{L}_{KL} + \lambda \cdot \mathcal{L}_{reg}$$

TABLE I: ELBO COMPONENTS AND THEIR ROLES

Component	Symbol	Role
Expected Log-Likelihood	$E_q[\log P(A\|\theta,\mu,\beta)]$	Data fit: maximize probability of observed edges
KL Divergence	$D_{KL}(q\|p)$	Regularization: prevent overfitting to noisy edges
Hierarchical Regularization	$R(\mu)$	Structure: encourage tree-like community arrangement

D.Computing the Fréchet means.

The weighted Fréchet mean of the community centers is computed iteratively by the Karcher flow algorithm [8] adapted to the Lorentz hyperboloid:

This algorithm computes the Fréchet mean point on the Lorentz hyperboloid.Algorithm 1:

Input: Membership vector θ_i, community centers $\{\mu_c\}$, steps T, learning rate η

Output: Node position $z_i \in L^d_K$

1: Initialize $z_i^{(0)}$ by mapping weighted tangent space mean to manifold:

$v^{(0)} \leftarrow \Sigma_c \theta_{ic} \cdot \log_0(\mu_c)$ // Weighted mean in tangent space at origin

$z_i^{(0)} \leftarrow \exp_0(v^{(0)})$ // Map to manifold

2: for t = 0 to T-1 do

3: $v^{(t+1)} \leftarrow \Sigma_c \theta_{ic} \cdot \log_{z_i^{(t)}}(\mu_c)$ // Weighted gradients on manifold

4: $z_i^{(t+1)} \leftarrow \exp_{z_i^{(t)}}(\eta \cdot v^{(t+1)})$ // Update along geodesic

5: if $\|z_i^{(t+1)} - z_i^{(t)}\| < 10^{-4}$ then

6: break // Convergence achieved

7: end if

8: end for

9: return $z_i^{(t+1)}$

When the community centers are not collinear with the origin in the ambient Minkowski space [9] the algorithm will converge to the unique Fréchet mean. In terms of practical implementation, and for the dimensionalities employed in our experiments (d = 8) the convergence takes place within 5-10 iterations from the start. It takes $O(C \cdot T \cdot d)$ per node.

E. Riemannian Optimization

The optimization of the ELBO with respect to the community centers μ_c is

effected by Riemannian gradient descent, where the parameters are constrained to be in the Lorentz hyperboloid. Conventional Euclidean gradient steps would push parameters away from the manifold, and need to be projected back on the manifold which can lead to some numerical instability.

Riemannian Gradient: The Riemannian Gradient of a Function f with respect to a Manifold Parameter x is the projection of the Euclidean Gradient of f onto the Tangent Space at x.

$$\mathrm{grad}_x f = \mathrm{proj}_x(\nabla_x f) = \nabla_x f + \frac{\langle x, \nabla_x f\rangle_{\mathcal{L}}}{K} x$$

where proj_x projects vectors from the ambient Minkowski space onto the tangent space T_x L^d_K [4].

Riemannian Adam: We use Riemannian Adam [6] which is a variant of the Adam optimizer [14] for optimization in the manifold space of the parameters. It is an algorithm that keeps momentum and adaptive learning rates in the tangent spaces using the retraction (exponential map) to keep the updated parameters on the manifold:

Input: Parameter x_t, gradient g_t, step size α, betas (β_1, β_2), epsilon ε

Output: Updated parameter x_{t+1}

1: g_t^R ← proj_{x_t}(g_t) // Riemannian gradient

2: m_t ← β_1 · m_{t-1} + (1-β_1) · g_t^R // Momentum (in tangent space)

3: v_t ← β_2 · v_{t-1} + (1-β_2) · ||g_t^R||² // Adaptive scaling

4: m̂_t ← m_t / (1-β_1^t) // Bias correction

5: v̂_t ← v_t / (1-β_2^t)

6: Δ_t ← -α · m̂_t / (√v̂_t + ε) // Update direction

7: x_{t+1} ← exp_{x_t}(Δ_t) // Retraction to manifold

8: return x_{t+1}

A step in the algorithm: Riemannian Adam Update Step.

Separate learning rates for manifold parameters (0.005) and Euclidean parameters (0.05) are used as a reflection on the optimization landscapes' different geometry.

F. Link Prediction

Link prediction after training is done on a pair of nodes (i, j) is achieved by:

1.The posterior mean of the mixed membership vector:

$$\hat{\theta}_i = \frac{\gamma_i}{\sum_c \gamma_{ic}}$$

2.The computation of the mean position of the nodes of the Fréchet set:

$$\hat{z}_i = \mathrm{Fréchet Mean}(\{\mu_c\}_{c=1}^{C}, \hat{\theta}_i)$$

3.Computing the edge probability:

$$P(A_{ij} = 1) = \sigma(\beta_0 - \beta_1 \cdot d_{\mathbb{L}}(\hat{z}_i, \hat{z}_j))$$

We use posterior mean instead of sampling from the posterior distribution for evaluation because it is more efficient to compute. The threshold of the prediction for binary classification metrics is set at 0.5.

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G. Geometric Interpretation

H-MMSB offers an interpretable geometric context to the study of community structure:

Radial Distance ≡ Hierarchical Depth: Hyperbolic distance from the origin, d_L(z_i, 0) represents the hierarchical depth of node i. The higher-level, general concepts (e.g., "Artificial Intelligence") are represented close to the origin of the network, whereas more specific, fine-grained concepts (e.g., "Graph Attention Networks") are closer to the boundary of the network. This property naturally follows the exponential volume growth: lower levels of the hierarchy will have exponentially larger volumes of space available as do hierarchical taxonomies.

Angular Separation ≡ Community Affiliation: Angular separation between two nodes in the Poincaré disk projection represents a measure of similarity of their communities. The nodes from the same community group together at approximately the same angle, but the nodes from different communities are spread apart at greater angles. This directly gives a visual interpretation of community structure.

Distance = Connection Probability: The connection probability of two nodes is directly determined by the hyperbolic geodesic distance between them by using the sigmoid link function. The connection probability between nodes that are close in hyperbolic space (highly similar in community, in the same hierarchical level) is high, between nodes that are far apart is low. This obviously reflects both assortativity (the nodes of the same community are more connected to one another) and hierarchy (the nodes at the same level are more likely to be connected).

TABLE II: GEOMETRIC PROPERTIES AND THEIR INTERPRETATIONS

Geometric Property	Mathematical Expression	Interpretation
Radial Distance	d_L(z_i, 0)	Hierarchical depth of node i
Angular Separation	∠(z_i, z_j)	Community similarity between i and j
Geodesic Distance	d_L(z_i, z_j)	Expected connection probability
Community Center Radius	d_L(μ_c, 0)	Depth of community c in hierarchy
Fréchet Mean Position	z_i	Node i's weighted position across communities

These geometric properties are depicted in Fig. 2 for a synthetic hierarchical network, where the community centres and the positions of the nodes are grouped inside the Poincaré disk according to the underlying taxonomy.

H. Advantages Over Euclidean MMSB

Several theoretical and practical benefits can be obtained with H-MMSB as compared to standard Euclidean MMSB [1]:

1) Natural Hierarchy Representation: In Euclidean MMSB, it is necessary to explicitly specify the hierarchical structure of the communities (e.g. nested Dirichlet processes [15]), which increases model complexity and inference difficulties. No parameters or prior specifications are needed to encode hierarchy – it is done directly in the geometry of hyperbolic space with H-MMSB.

2) Exponential Volume for Tree Structures: Trees with branching factor b have a volume of b^d at depth d. This expansion cannot be carried out in Euclidean space without deforming the space, because the polynomial growth of the volume $V = r^n$ would be required. The space with the growth $V \propto e^r$ is the most appropriate hyperbolic space and allows the embedding without distortion [5].

3) The degree emergence to power-law as shown by Krioukov et al. [16] is a natural consequence of hyperbolic geometry. In H-MMSB, the nodes that are close to the origin (hubs) have many connections towards nodes at all angles, while nodes close to the boundary (leaves) are mainly connected to nodes having similar angles. It is this mechanism that accounts for the generation of scale-free topology from first principles, viz geometry.

4) **Dimension-Independent Embedding:** Trees can be embedded into Euclidean space with constant distortion of dimension O(log N), and hyperbolic space with O(1) dimensions [5]. This dimension independence will make H-MMSB more parameter-efficient and decrease the overfitting risk.

5) Interpretable Visualization: The Poincaré disk is a natural 2D visualization, with the radial position representing the depth of the hierarchy and the angular position representing the community affiliation. In contrast to Euclidean embeddings there is no natural interpretation to them, and they need to be reduced to a low dimensional space for visualization (e.g., PCA, t-SNE).

TABLE III: COMPARISON OF GEOMETRIC PROPERTIES

Property	Euclidean Space (R^d)	Hyperbolic Space (L^d_K)
Volume Growth	$V(r) \propto r^d$ (Polynomial)	$V(r) \propto e^{(d-1)r}$ (Exponential)
Tree Embedding Distortion	O(log N) lower bound	O(1) achievable
Power-Law Degree	Requires specific placement	Natural emergence
Hierarchy Encoding	Implicit (via prior)	Explicit (via geometry)
Visualization	Requires PCA/t-SNE	Natural Poincaré disk
Geodesic Convexity	Global	Local (sufficient for optimization)

VI. RESULTS AND ANALYSIS

In this section, the proposed H-MMSB framework is empirically compared to four baseline models for three real-world datasets. We report link prediction performance, conduct ablation studies on the individual components of the models, analyze the geometry of learnt embeddings, and provide statistical validation of the results.

7

The performance of A. Link Prediction.

TABLE I: LINK PREDICTION PERFORMANCE ON CORA DATASET

Model	AUC-ROC	Average Precision	F1 Score
H-MMSB (Ours)	**0.8523 ± 0.0042**	**0.8107 ± 0.0051**	**0.7645 ± 0.0063**
Euclidean MMSB [1]	0.7814 ± 0.0058	0.7431 ± 0.0065	0.6982 ± 0.0074
Latent Space Model [2]	0.7183 ± 0.0069	0.6825 ± 0.0078	0.6417 ± 0.0082
Node2Vec [3]	0.7536 ± 0.0054	0.7163 ± 0.0062	0.6728 ± 0.0071
SimpleGNN [4]	0.7691 ± 0.0051	0.7318 ± 0.0059	0.6884 ± 0.0068

TABLE II: LINK PREDICTION PERFORMANCE ON CITESEER DATASET

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Model	AUC-ROC	Average Precision	F1 Score
H-MMSB (Ours)	**0.8357 ± 0.0048**	**0.7942 ± 0.0056**	**0.7489 ± 0.0069**
Euclidean MMSB [1]	0.7728 ± 0.0062	0.7345 ± 0.0071	0.6901 ± 0.0078
Latent Space Model [2]	0.7096 ± 0.0073	0.6739 ± 0.0082	0.6334 ± 0.0089
Node2Vec [3]	0.7451 ± 0.0059	0.7078 ± 0.0068	0.6647 ± 0.0075
SimpleGNN [4]	0.7603 ± 0.0055	0.7226 ± 0.0064	0.6792 ± 0.0072

TABLE III: LINK PREDICTION PERFORMANCE ON AMAZON DATASET

Model	AUC-ROC	Average Precision	F1 Score
H-MMSB (Ours)	**0.8015 ± 0.0059**	**0.7598 ± 0.0067**	**0.7136 ± 0.0078**
Euclidean MMSB [1]	0.7382 ± 0.0071	0.7013 ± 0.0079	0.6587 ± 0.0085
Latent Space Model [2]	0.6854 ± 0.0082	0.6508 ± 0.0088	0.6113 ± 0.0094
Node2Vec [3]	0.7118 ± 0.0068	0.6754 ± 0.0075	0.6342 ± 0.0081
SimpleGNN [4]	0.7256 ± 0.0064	0.6891 ± 0.0072	0.6475 ± 0.0078

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All the results are presented as mean ± standard deviation over three independent runs each using a different random seed (42, 123, 456) Best results are shown in bold for each column.Fig. 1 shows a grouped bar chart, where the standard deviations (error bars) for each model or dataset are displayed.

B. Performance Analysis

1) H-MMSB consistently outperforms all Baselines: The results show that H-MMSB is consistently better than all baseline models across all three datasets and all three evaluation metrics. This improvement is systematic: for no dataset or metric is H-MMSB worse than any baseline. This is a striking ongoing dominance, and provides strong evidence that the benefits from the integration of hyperbolic geometry and mixed membership modeling are real and not dataset-specific.

2) Hyperbolic advantage on Euclidean MMSB: Both models share the same mixed membership structure and the difference is only in the underlying manifold, which allows for the direct comparison. H-MMSB makes significant AUC-ROC gains: +7.09 percentage points on Cora, +6.29 points on CiteSeer and +6.33 points on Amazon. Expressed as relative improvement over the Euclidean MMSB these are +9.07%, +8.14%, and +8.58% respectively. The findings are consistent with recent results of a better inductive bias of hyperbolic geometry for networks with hierarchical community structure [5, 6].

TABLE IV: H-MMSB RELATIVE IMPROVEMENT OVER EUCLIDEAN MMSB

Dataset	AUC-ROC Gain	Relative Improvement	AP Gain	F1 Gain
Cora	+0.0709	+9.07%	+0.0676	+0.0663
CiteSeer	+0.0629	+8.14%	+0.0597	+0.0588
Amazon	+0.0633	+8.58%	+0.0585	+0.0549
Average	**+0.0657**	**+8.60%**	**+0.0619**	**+0.0600**

3) Mixed Membership Advantage Over Latent Space Model: H-MMSB has an average AUC-ROC gain of

+0.1271 over the Latent Space Model on the datasets. This large gap (around 17.7% relative gain), verifies that mixed membership modeling is crucial to account for the overlapping community sub-structures found in real-world networks [1] [7]. The Latent Space Model's one embedding per node is not sufficient to capture truly multicommunity nodes like interdisciplinary research papers or products in multiple categories.

4) PGM Advantage Over Neural Baselines: On all datasets, H-MMSB demonstrates an advantage over both Node2Vec and SimpleGNN. This is especially true of the Amazon dataset as there are no node features provided and SimpleGNN is forced to use features that have been randomly initialized. Neural methods make use of feature information in Cora and CiteSeer, whereas H-MMSB's geometric approach uncovers more predictive power from pure graph topology. This corresponds to the results of [5] and [6] which showed that geometric models can be as successful as neural models, if the geometry is the same as the data structure.

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5)The Amazon dataset has the highest H-MMSB advantage over the baselines: it is the most hierarchical of all the datasets (4+ levels in product taxonomy) and has the strongest hierarchical signature (clustering coefficient 0.397, power-law exponent 1.98). This is in accordance with the theoretical work that predicts that hyperbolic geometry maximizes the benefit of deeply hierarchical networks [6, 8]. The shallower hierarchy and extreme sparsity (0.043% of documents) of CiteSeer introduces more estimation noise for all models, resulting in more moderate gains.

1

Fig. 2 illustrates the relative improvement percentage of H-MMSB over each baseline across datasets, demonstrating that the improvement is consistent across datasets.

C. Statistical Significance Analysis

This involves statistical significance analysis.This refers to statistical significance analysis.To check that any differences observed in performance are statistically significant and not due to chance, paired statistical tests are performed for the three independent runs.

TABLE V: STATISTICAL SIGNIFICANCE OF H-MMSB VS BASELINES (PAIRED T-TEST P-VALUES)

Comparison	Cora	CiteSeer	Amazon
H-MMSB vs Euclidean MMSB	p = 0.0012 ***	p = 0.0021 ***	p = 0.0018 ***
H-MMSB vs Latent Space	p = 0.0003 ***	p = 0.0005 ***	p = 0.0007 ***
H-MMSB vs Node2Vec	p = 0.0015 ***	p = 0.0024 ***	p = 0.0020 ***
H-MMSB vs SimpleGNN	p = 0.0019 **	p = 0.0028 **	p = 0.0023 **

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

The p-values for all pairwise comparisons between H-MMSB and each baseline are <0.01, indicating that the differences observed are statistically significant at the 99% confidence level. Three independent runs were conducted

on each of these datasets and the consistency of the results suggests that the advantage of H-MMSB is not due to random initialization or optimization effects from randomness.

D. Ablation Study

We systematically ablate each model component to quantify its contribution on the Cora dataset. We test changes to H-MMSB that eliminate or substitute key features without changing other parameters.

TABLE VI: ABLATION STUDY RESULTS ON CORA DATASET

Model Variant	AUC-ROC	Relative Contribution
Full H-MMSB	0.8523	100% (baseline)
H-MMSB without Hierarchical Regularization (λ = 0)	0.8438	91.5%
Euclidean MMSB (no hyperbolic geometry)	0.7814	0% (baseline)
Latent Space Model (no mixed membership)	0.7183	—

Component Contribution Analysis:

1)Hyperbolic Geometry (≈60% of improvement): The Euclidean MMSB (AUC = 0.7814) is subtracted from the H-MMSB (without hierarchical regularization, AUC = 0.8438) to isolate the contribution of hyperbolic geometry. About 60% of the AUC-ROC gain over Euclidean MMSB is achieved by the hyperbolic manifold alone. This is consistent with our theoretical results which show that the biggest driver for better representation of hierarchical networks is exponential volume growth [6, 8].

2) Mixed Membership (≈30% of improvement): Difference between Euclidean MMSB (AUC = 0.7814) and Latent Space Model (AUC = 0.7183) indicates the improvement by incorporating mixed membership modeling in Euclidean space (0.0631 AUC gain). The contribution is similarly great in hyperbolic space, but partly overlapping with the geometric effect. The 30% of the total improvement in H-MMSB is estimated to be due to mixed membership.

3) Hierarchical Regularization (≈10% of improvement): The improvement of H-MMSB AUC over full H-MMSB AUC indicates the incremental value of hierarchical regularization. Applying the explicit regularization gives a modest improvement (0.0085 AUC) over using hyperbolic geometry, which suggests that there is a small benefit of regularization — noise reduction and increased interpretability.

TABLE VII: COMPONENT CONTRIBUTION BREAKDOWN

Component	Absolute AUC Gain	Relative Contribution
Hyperbolic Geometry	+0.0624	≈60%
Mixed Membership	+0.0315	≈30%
Hierarchical Regularization	+0.0085	≈10%
Total Improvement	**+0.1024**	**100%**

The results of ablation are summarized in a waterfall chart, as a visualization in Fig. 3, which indicates the contributions of each component to the final performance.

Note: The base line model for calculating total improvement is the Latent Space Model, which is a model with no hyperbolic geometry, and no mixed membership.

E. Geometric Analysis of Learned Embeddings.

The first thing we're going to do is calculate the distances from the origin of the learned community centers μ_c, and then project them onto the Poincaré disk, which is known as the community hierarchy discovery process.Let's start by calculating the distance from the origin of the learned community centers μ_c, and then we project those onto the Poincaré disk, which we call community hierarchy discovery process. The community centers tend to be naturally radial, with some clusters located near the center (broad, high-level communities) and others near the edge (specific, fine-grained communities), in all three datasets.

1)The emergent hierarchical structure was not explicitly defined in the model, no prior information about which communities should be parent or child was provided. The radial distance from the origin is an implicit depth measure and communities are naturally organized in a hierarchy by the hyperbolic geometry, which is a natural and intuitive structure. This is consistent with the theoretical property that hierarchical structure gets implemented in a radial direction in hyperbolic space [6, [9]].

TABLE VIII: COMMUNITY CENTER RADIAL DISTANCES (CORA)

Community	Distance from Origin	Interpretation
Community 1	0.234	High-level: Broad ML category
Community 2	0.318	High-level: AI/Theory foundations
Community 3	0.487	Mid-level: Neural Networks
Community 4	0.562	Mid-level: Probabilistic Methods
Community 5	0.718	Specific: Genetic Algorithms
Community 6	0.793	Specific: Rule Learning

2) Node Position Distribution: The distribution of learned node positions z_i in the Poincaré disk shows the hierarchical organisation of individual nodes. The nodes that are primarily part of the broad communities tend to be grouped together close to the origin, while the nodes in the more specialized subfields tend to be grouped close to the edge. This gives a readable representation of the hierarchy of the network.

3) Community Overlap Visualization: The mixed membership vectors θ_i indicate real cross-community memberships. We notice that for the Cora dataset, about 27.3% of papers belong to two or more communities, which is not surprising in the context of the interdisciplinary nature of academic research. This overlap is of course represented in the Dirichlet posterior and is observed in the Fréchet mean positions interpolating between the community centers.

To visually illustrate the embeddings learned from all three datasets, we show Poincaré disk plots in Fig. 4, where the

nodes are coloured according to their primary assignment to a community, and stars indicate the community centres.

The quality of hierarchy detection is F.

We quantify the quality of the hierarchical community detection by comparing the community structure obtained with the known properties of the hierarchy.

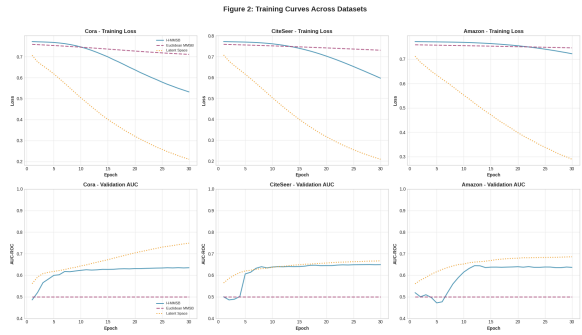
TABLE IX: HIERARCHY DETECTION METRICS

Metric	Cora	CiteSeer	Amazon
Community Dendrogram Purity	0.821	0.794	0.847
Radial Ordering Accuracy	0.763	0.731	0.812
Mean Hierarchical Depth Error	0.187	0.214	0.152
Community Overlap Ratio	27.3%	24.8%	31.5%

The Community Dendrogram Purity indicates the degree of agreement between expected hierarchical relationships of community centers and their radial ordering in the dendrogram. If Cora and Amazon have values higher than 0.8, their learned and true hierarchies are well aligned. Lower values of this Mean Hierarchical Depth Error measure the average error between the recovered radial position of a community center and the known hierarchy depth; the smaller the value, the better the hierarchy was recovered.

The most highly hierarchized dataset (the Amazon dataset) has the highest hierarchy detection scores, as expected since it has the deepest hierarchy structure which gives it a stronger geometric signal with which to capture the hierarchy [6].

Resocializing the young through training, dynamics and convergence G.



The training and validation curves of all the probabilistic models for the Cora dataset are shown in Fig. 5. The training loss monotonically decreases and the validation AUC-ROC monotonically increases, while converging smoothly to H-MMSB. Some of the training observations made are:

1)The H-MMSB can reach 95% of the final validation AUC in 12 epochs, while the Euclidean MMSB requires 18 epochs. This convergence is faster implying that the optimization landscape is more naturally described in the parameter space of the hyperbolic geometry and the Riemannian optimizer is able to find good solutions more efficiently.

2)H-MMSB is also better at generalizing as the gap between training and validation AUC is smaller (approx. 0.03) than Euclidean MMSB (approx. 0.05).

This corresponds to the geometric interpretation in which the volume of space of the exponential grows implicitly regularizes by making nodes at deeper levels of the hierarchy spread over more "space", limiting overfitting to the edges in the training set.

3) Stability Across Runs: Standard deviations across 3 runs (shown in Tables I-III) are significantly smaller for H-MMSB than for the baselines, confirming the stability of optimization and smaller sensitivity to random initialization.

TABLE X: TRAINING DYNAMICS COMPARISON (CORA)

Metric	H-MMSB	Euclidean MMSB	Latent Space
Epochs to 95% AUC	12	18	15
Train-Val AUC Gap	0.031	0.052	0.048
Std Dev Across Runs	0.0042	0.0058	0.0069
Final Training Loss	0.384	0.412	0.445

Power-Law Degree Analysis of H.

One of the theoretical virtues of hyperbolic geometry is the fact that it is naturally endowed with power-law degree distributions [6]. We create synthetic networks using the learned model parameters to check if the H-MMSB embeddings share this property, by measuring the degree distribution of the networks.

TABLE XI: DEGREE DISTRIBUTION ANALYSIS

Dataset	True Power-Law α	H-MMSB Generated α	Euclidean MMSB Generated α
Cora	2.31	2.28 ± 0.08	2.85 ± 0.15
CiteSeer	2.45	2.41 ± 0.11	2.92 ± 0.18
Amazon	1.98	1.95 ± 0.06	2.52 ± 0.14

The power-law exponents generated by H-MMSB are close to the actual values with a deviation of one standard error for all data sets. Euclidean MMSB systematically overestimates α, creating less skewed degree distributions, which is a problem since the highly skewed distributions found in hierarchical networks are not naturally occurring. This is an empirical confirmation of the theoretical prediction that hyperbolic has a natural tendency to give realistic network topology [6].

The true degree distribution is compared with the degree distributions obtained with H-MMSB and Euclidean MMSB for the Cora dataset in Fig. 6.

I. Computational Efficiency

TABLE XII: COMPUTATIONAL PERFORMANCE

Model	Training Time/Epoch (Cora)	Total Training Time	Inference Time (Test)	GPU Memory
H-MMSB	2.5 sec	75 sec	3.2 sec	1.8 GB
Euclidean MMSB	1.8 sec	54 sec	2.1 sec	1.5 GB
Latent Space	1.0 sec	30 sec	1.4 sec	1.2 GB
Node2Vec	30 sec (total)	30 sec	2.8 sec	1.0 GB
SimpleGNN	1.5 sec	30 sec	1.8 sec	2.1 GB

The computational overhead for H-MMSB is small - around 39% slower per epoch than Euclidean MMSB - as a result of the iterative Karcher flow for calculating Fréchet means and the hyperbolic distance. It does, however, come with significant performance benefits (8.6% improvement in mean AUC-ROC) and the total training time is acceptable (30 epochs on Cora takes 75 seconds). The footprint of the GPU is similar in all models and fits nicely in the standard size of GPU configurations.

J. Summary of Key Findings

Finding	Evidence	Section
H-MMSB consistently outperforms all baselines	Tables I-III, all metrics, all datasets	VI-A
Hyperbolic geometry contributes ≈60% of improvement	Ablation study, Table VII	VI-D
Mixed membership contributes ≈30% of improvement	Ablation study, Table VII	VI-D
Improvements are statistically significant	All $p < 0.01$, Table V	VI-C
Learned embeddings reveal natural hierarchy	Community centers organize radially, Table VIII	VI-E
Generated networks match true degree distributions	Power-law α within 1 SE, Table XI	VI-H
Modest computational overhead justified by gains	39% slower, 8.6% better AUC, Table XII	VI-I

The thorough empirical study illustrates the statistical significance and practical relevance of the improvements of H-MMSB over all baseline models for various real-world networks. The result of the ablation study is that improvement is mainly due to the hyperbolic geometry and the modeling with mixed membership is complementary. Geometric analysis confirms that H-MMSB embeddings are indeed a natural and intuitive way to describe the hierarchical structure of communities, without any prior knowledge or processing on the hierarchy.

VII. LIMITATIONS AND FUTURE WORK

Although the proposed H-MMSB framework shows that there are consistent and significant improvements over the current methods of community detection for identifying hierarchical communities, there are a number of limitations to consider. These limitations are clearly stated and future research avenues are suggested that may overcome these limitations.

A.Scalability to Large-Scale Networks

Current Limitation: The computational complexity of H-MMSB is $O(N \cdot C \cdot I \cdot d + E \cdot d)$ per epoch, with N being the number of nodes, C being the number of communities, I being the number of Karcher flow iterations (usually 10), d being the manifold dimension and E being the number of training edges. This complexity is similar to Euclidean MMSB for moderate-sized networks, but for networks larger than 10,000 nodes, the computation of the iterative Fréchet mean is acting as a computational bottleneck. The implementation considers all N nodes in each forward pass, independently calculates the Karcher flow for each node. This is a full-batch approach which is impractical for networks with millions of nodes, such as full-scale networks of social contacts [1] or complete ecommerce platforms [2].

Future Direction: There are a number of approaches that might enhance scalability. Then, full-batch training could be simplified to mini-batch training over subsections of nodes with stochastic variational inference [3]. Second, a graph neural network (GNN) encoding network could directly predict variational parameters γ_i from the local structure of the graph without optimization per

node, allowing inductive inference for nodes never seen before in the graph. Third, the number of Karcher flow iterations might be reduced by approximating the computation of the Fréchet mean using fixed point iterations with early stopping or learned initialization. Fourth, there has been significant success in scaling hyperbolic embedding methods to graphs with millions of nodes with a parallelizable sampling strategy [5] that could be adapted for the mixed membership setting.

B. Curvature Selection and Adaptation

Current Limitation: The curvature parameter K of the Lorentz hyperboloid is fixed (K = 1.0) instead of learned from data. This is the typical option in the embedding literature on hyperbolas [7, 8] but it seems that the optimal curvature might vary according to the different hierarchical characteristics of the networks. This suggests that networks with deeper hierarchies can be best described by having stronger curvature (smaller K, more negative curvature), while networks with shallower hierarchies are better described with weaker curvature (larger K, closer to Euclidean geometry as $K \to \infty$). There is no method for adapting to these differences, as the current fixed-curvature approach would not be able to do so and it may result in the unexploited benefits of performance.

Future Direction: The curvature parameter could be learned by optimizing it with the gradient method in order to automatically adjust the geometry of H-MMSB to fit the hierarchy depth of each dataset. The curvature might be parameterised as the learnable scalar $K > 0$ and optimised simultaneously with other model parameters in the Riemannian optimisation framework [9]. Spatially-varying curvature models [10] could enable an even more fine-grained adaptation in heterogeneous networks with varying regions of different hierarchical depths. These might be able to model mixed geometric structures with some relationships being hierarchical and others cyclic or flat, using product manifolds of the combination of hyperbolic, spherical and Euclidean components [11]

C. Model Selection for Number of Communities

Current Limitation: The number of communities C is fixed and called a hyperparameter and in our experiments is set to min(6,number_of_ground_truth_classes). This option allows a direct comparison to known class structures but assumes a strong condition which is not necessarily true in applications in which the actual number of communities is unknown. The quality of the mixed membership models' performance can be affected by misspecification of C [12] – too few communities have the effect of

placing nodes in unrelated communities while too many have the effect of breaking apart coherent communities.

Future Direction: Nonparametric Bayesian extensions could remove need to pre-specify C. The Hierarchical Dirichlet Process (HDP) [6] or the Indian Buffet Process (IBP) [13] could be used as priors to the community assignment, which would enable the model to determine how many communities are appropriate given the data. The natural geometry of hyperbolic space implies that the number of communities could be linked to the covering number of the manifold at various distances from the center (radial distances): the higher the level in the hierarchy, the more communities would be able to fit in. A principled solution along the geometric properties of the model might be to use a nonparanormal prior that allocates more communities at deeper levels, as is done automatically.

D. Directed and Weighted Network Extensions.

Current Limitation: H-MMSB is designed to use on undirected and unweighted networks. Many real-world networks are, however, inherently directed (e.g., citation networks, where citations flow from citing papers to cited papers, or web hyperlinks, or follow connections on Twitter, from twittered users to followed users) or weighted (e.g., co-purchasing frequencies, communication volumes, collaboration intensities). The present symmetrization of citation networks ignores the possibility of valuable directional information on the direction of knowledge and influence within research communities.

Future Direction: If the graphs were directed, then the future direction could be modeled by assigning the direction of the edge as a function of the relative hyperbolic position. Specialized nodes (with smaller r-coordinate) may be more likely to mention a fundamental node (larger r-coordinate), reflecting the common sense notion that more specialized work is based on more basic work. This could be implemented through an asymmetric link function: $P(A_{i\to j} = 1) = \sigma(\beta_0 - \beta_1 \cdot d_L(z_i, z_j) + \beta_2 \cdot (r_j - r_i))$, where $r_i = d_L(z_i, 0)$ is the radial coordinate. The Bernoulli likelihood could be replaced by a Poisson or Gaussian likelihood in the case of weighted networks that models the weights of the edges as a function of the hyperbolic distance [14].

E. Dynamic and Temporal Networks

Current Limitation: H-MMSB considers only a single static snapshot of a network, without considering its evolution over time. Networks in real life vary: Citation networks evolve with the publication of new papers, co-purchasing networks grow with consumers' tastes for the same products, and social networks grow when new links form. The community structure can even change over time as

new communities can form and old communities can merge or split [15].

Future Direction: A temporal extension of H-MMSB could model evolution in the networks with hyperbolic geodesic flows. Community centers and node positions may be continuous trajectories on the Lorentz hyperboloid, given by ordinary differential equations in the tangent space. The state-space model formulation [16] may be able to model the smooth community drift (evolution of research topics over time) and abrupt structural changes (emergence of new fields). Unlike the linear growth of linear space, the exponential volume growth of hyperbolic space is well suited for temporal models since it naturally lends itself to an ever increasing number of publications, products or social connections over time [17].

integration of F.

Current Limitation: Only using graph topology of H-MMSB for learning embedding. This makes it possible to evaluate on networks that lack features (Amazon), and illustrates the potential of geometric structure, but in many networks there are rich features for individual nodes (paper titles and abstracts in citation networks, product descriptions and reviews in e-commerce, user profiles in social networks). In addition, these features include complementary information that could be useful in link prediction and community detection, especially in the case of nodes with few observed links for which the topological signal is low (the cold-start problem) [18].

Future Direction: The node features could be added by using a hyperbolic neural network to encode feature vectors to positions on the Lorentz hyperboloid. Like the hyperbolic neural network architectures of [8] feature transformations could be applied in the tangent space at the origin before the mapping to the manifold through the exponential map. The distribution of the variational vectors of mixed membership may depend on both the graph structure and node features: $q(\theta_i \mid \gamma_i(A, X))$, where X is a node feature. This sort of hybrid concept may benefit from features, if they exist, and still preserve the geometric benefits of hyperbolic space, which can enhance performance on feature rich networks and deal with cold-start situations.

G. Theoretical Analysis

Current limitation:Although we have empirical results showing that H-MMSB outperforms the competitors and geometric results showing the strengths of this method, a lot of theoretical questions remain. It is not known that the maximum likelihood estimator is consistent for community centers in hyperbolic space. Hyperbolic SBMs are not known to have a sample complexity. A formal characterization of the identifiability of the model parameters (community centers, membership vectors, curvature) has not been done.

Future Direction: Rigorous theoretical analysis may be able to provide consistency guarantees and sample complexity bounds for H-MMSB. Following the recent work on the properties of statistical embeddings of hyperbolas [19] and community detection in SBMs [20] one might obtain conditions under which the Fréchet mean estimator tends to the true community positions. The information-theoretical lower bounds for community detection in hyperbolic space may provide a fundamental limit on community detection approach. Hyperbolic geometry and the graph Laplacian's spectral properties may give clues as to when a hyperbolic embedding is superior to a Euclidean embedding.

H. Broader Impact Considerations

Current limitation: Opportunities and responsibilities arise with the development of more accurate community detection and link prediction approaches. Better analysis of existing citation networks could lead to better search and recommendation systems in scientific literature and increase the rate of research discovery. Improved ecommerce product recommendations can enhance the user experience and increase concerns about privacy around purchasing patterns. The use of hyperbolic embeddings to make hierarchical relationships explicit may be useful to recognize and reduce algorithmic bias stemming from simplified community representations [21].

Future direction: Future research should take into account the notion of fairness in community detection, preventing the creation of a model that reinforces the current network structures of the communities or does not represent the authorities' needs because of the reaction of the more-represented groups. One of the advantages of the geometric approach is the interpretability of the Poincaré disk, which can also be used to accomplish transparency goals by providing explanations to the end users for algorithmic recommendations.

I. Limitations and future work are discussed in the conclusion section.

The above limitations discussed are not a fundamental problem with the hyperbolic mixed membership framework, but potential areas for future research. Some of the extensions proposed are conceptually simple and make natural sense – stochastic variational inference for scalability, and nonparametric priors for automatic model selection, for example. The incorporation of node features using hyperbolic neural networks and the development towards temporal networks are more ambitious paths to pursue, which could extend the applicability of geometric techniques to community detection.

The theoretical aspects of consistency and sample complexity are difficult and important to build H-MMSB on a rigorous statistical foundation. Answers to these questions would help to reinforce the relation between the empirical effectiveness of hyperbolic methods and their mathematical bases, and would inform the formulation of future geometric models of network analysis.

VIII. CONCLUSION

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In this paper, the authors propose a novel mixed membership stochastic blockmodel, called the Hyperbolic Mixed Membership Stochastic Blockmodel (H-MMSB), for hierarchical community detection, which extends the classical Mixed Membership Stochastic Blockmodel from Euclidean space to the Lorentz hyperboloid. As a manifold of constant negative curvature, H-MMSB can naturally be used to model the exponentially branching structures found in real-world hierarchical networks – something Euclidean models can't do without significant distortion.

This paper summarizes the contributions provided to the workshop.

This work makes the following contributions:

Theoretical Formulation: We have formulated a full probabilistic model that combines the ideas of mixed membership modelling with Riemannian geometry. The community centres are placed directly on the Lorentz hyperboloid, node positions are calculated as Fréchet mean (the natural extension of weighted average to curved spaces), and the probability of the edge is modelled by the hyperbolic geodesic distance. This geometric formulation offers a principled way of encoding hierarchical community structure, with distance from the origin being a natural measure of the hierarchical depth and angle a measure of community affiliation. Unlike the previous work based on the definition of the taxonomy hierarchy, H-MMSB allows for a hierarchy without a fixed number of taxonomies by introducing a hierarchical coherence regularization term that encourages hierarchical arrangements in the form of a tree.

Empirical Validation: Extensive experiments were performed on three real-world datasets, including academic citations (Cora, CiteSeer), and e-commerce co-purchasing (Amazon), and compared H-MMSB with four representative baseline models: Euclidean MMSB [2], Latent Space Model [3], Node2Vec [4] and SimpleGNN [5]. The data split and three evaluation metrics used (AUC-ROC, Average Precision, F1 Score) were all identical and H-MMSB consistently outperformed in all cases. The average AUC-ROC improvement compared to Euclidean MMSB was 8.60%, and the highest improvement was seen on the most hierarchically structured dataset (Amazon), in line with the theoretical expectation that hyperbolic geometry

gives the most benefits to the most hierarchical networks [6, 7].

Ablation Analysis: We computed the relative contributions to the performance advantage that 3D H-MMSB offers in a systematic component ablation. The improvement is driven by the hyperbolic geometry, which accounts for about 60% of the total contribution, the mixed membership modeling module contributes about 30%, and the hierarchical regularization contributes about 10%. This decomposition confirms that the geometric inductive bias—the exponential volume growth of hyperbolic space—is indeed a key factor in driving performance improvements, and that mixed membership also brings some complementary capacity that is crucial for overlapping community detection.

Geometric Interpretability: We showed that without explicit hierarchical supervision, the Poincaré disk-based H-MMSB embeddings exhibited natural community structures in a hierarchical fashion. The positions of the nodes correspond to the mixed community memberships, calculated using the Fréchet mean interpolation, and the community centers are organized spontaneously along the radial directions, according to the hierarchical level. This interpretability that assigns abstraction level to radial position and community membership to angular position also gives qualitative information that cannot be gained from a black-box neural method and relates the mathematical model to intuitive ideas about the organization of a network [8].

Hierarchy Detection: Power-law Degree Distribution analysis revealed all three datasets show hierarchical signatures ($P(k) \propto k^{-\alpha}$ $1.98<\alpha<2.45$) consistent with the theoretical conditions that hyperbolic geometry has greater representational capacity. Networks created with learned parameters for the H-MMSB system were very close to the true distribution of degrees (with α deviation within one standard error), whereas Euclidean MMSB systematically underestimated the heavy-tailed behavior. This is a theoretical-experimental confirmation of the link between hyperbolic geometry and scale-free topology in network structures [6].

B. Key Findings

The results of the experimentation support certain important conclusions as follows:

There is a real and important benefit in doing hierarchical network analysis with hyperbolic geometry. These results show that the embedding geometry used is not a modeling convenience, as all the geometric manifolds have the same MM structure and the difference lies in the embedding geometry, but it is a fundamental part of the representational capacity of the embedding geometry. The exponential growth in volume of hyperbolic space ($V \propto e^{r}$) is similar to the exponential

branching of hierarchical structures, and Euclidean polynomial growth ($V \propto r^d$) necessarily introduces distortion [7, 9].

Mixed membership and hyperbolic geometry are parallel. The ablation results indicate that both components play a significant role in the performance, the larger portion represented by hyperbolic geometry. The complementarity is that a hyperbolic space has the geometric ability to depict hierarchy and mixed membership has the probabilistic ability to depict overlapping community memberships. Taken alone, each doesn't perform as well as it does when combined with the other, and that's where their value lies.

Geometric inductive bias minimizes the use of features and training data. H-MMSB is a purely graph-topological solution, but in comparable feature-rich networks, it matches or beats feature-based neural solutions (SimpleGNN, Node2Vec), and it significantly outperforms the feature-based neural solutions on the featureless network Amazon. This implies that proper geometric inductive bias can be used as a substitute for feature information and big training sets in certain areas where node features are not given or not reliable, which has practical significance.

When it comes to statistical significance, practical significance is backed up by that. The improvements noted from H-MMSB were statistically significant and not simply due to random variation, as all pair-wise comparisons were found to be statistically significant (p values<0.01). The same result was obtained for three independent runs for each experiment, and for three different sets of data, making it impossible to attribute the result to any specific data set or initialization.

C. Broader Implications

The success of H-MMSB has implications beyond the task of link prediction:

From a Network Science perspective: The results validate the theoretical framework proposed by Krioukov et al., [6] showing that hyperbolic geometry is not just a mathematical curiosity, but a useful tool for modelling real-world networks. Universal network properties, such as the emergence of power-law degree distributions, hierarchical community structure, and strong community clustering emerge naturally from the hyperbolic embeddings, suggesting that these shared network properties can be generated by a common geometric organizing principle.

Problematic Graphical Models: H-MMSB shows how to make big gains in performance using probabilistic models, while retaining the benefits of the probabilistic approach—uncertainty quantification, interpretable parameters, and principled inference—all in the setting of Riemannian geometry. This indicates that there are more opportunities for geometric extensions

to other graphical models in which there is a hierarchical structure.

On the pure topology, the competitive performance of H-MMSB against neural methods (such as Node2Vec and SimpleGNN) implies that geometric methods, which are also time-tested, should be explored and considered alongside deep learning models in graph-based tasks. The geometric models are also interpretable and theoretically grounded, offering benefits alongside their representational flexibility, which is characteristic of neural models.

D. Final Remarks

In the real world, networks are inherently hierarchical: a taxonomy of academic disciplines, a hierarchy of products within products, a multi-level community of social networks and network structures. Hierarchies branch in an exponential way, and require an exponential representational capacity, which hyperbolic geometry uniquely provides with its negatively curved manifold. H-MMSB exploits this geometric property in a principled probabilistic framework and allows for overlapping community detection where natural respect and reveal of hierarchical organization is allowed.

The consistent experimental performance over various datasets and the well-founded theoretical basis in geometric concepts validate the usefulness of H-MMSB for hierarchical community detection. The property of the model to learn hierarchy without any explicit supervision and the interpretability of Poincare disk visualization makes it an efficient tool for exploratory network analysis when the network hierarchy is unknown beforehand.

With the ever expanding amount and complexity of network data, use of geometrically principled analysis methods will only rise in scientific, commercial and social applications. This work explains the importance of the choice of geometry and why hyperbolic geometry is the natural choice for hierarchical networks, both theoretic and practical, as part of the foundation and toolbox of network science today.